

Exploring Educators Views and Experiences with Gender Inclusivity in Computer Science Higher Education

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A critical challenge in Computer Science Higher Education (CSHE) is gender disparity. Despite ongoing efforts, gender disparity remains a crucial challenge in CSHE, affecting student engagement and learning outcomes. We wanted to investigate CSHE educators' views on gender inclusivity or lack thereof, key factors influencing inclusivity in CSHE settings, and how various student facets, characteristics and behaviours, as perceived by education, influence different gender student participation and engagement. We used semi-structured interviews (n=20) with a range of experienced CSHE educators to explore these factors in depth. We report on our key findings, recommendations and future research directions.

CCS Concepts: • **Social and professional topics** → **Computer science education**; **Gender**.

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1 INTRODUCTION

A critical, yet pervasive issue in Computer Science (CS) education is gender disparity [39]. CS education and practice environments are still predominantly male-oriented [18]. Gender imbalance in CS education is a topic of scientific, industry and governmental research [25, 28, 48, 59]. Based on the current studies, approximately four out of every five students studying CS-related higher education courses in most countries are men [29, 49]. This uneven representation raises concerns regarding the barriers that women and other marginalised genders face in accessing and thriving in this field. Addressing this disparity requires investigating the existing causes and making a concerted effort to address them, particularly in the education domain.

Gender inclusivity in CS education has suffered for a long time due to cultural norms, pedagogical practices, institutional policies and both lack of enrolments and lower retention of non-male students [41]. Several studies have highlighted that CS Higher Education (hereafter referred to as CSHE) environments are often perceived as male-dominated, which can discourage women and other marginalized genders from participating fully in the field [33]. This is a global phenomenon and not restricted to any one country. Based on estimates in the US, female participation in CSHE, specifically, has declined to 20% from a 37% peak in mid 1980s [16]. Similar trends have been witnessed in other parts of the world, e.g., in Austria, Belgium, Denmark, Finland, Germany, Ireland,

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Italy, Latvia, Lithuania, the Netherlands, Poland, Spain, Switzerland, and the UK, 80% or more of the students enrolling or graduating in Informatics Bachelor programs are male [29]. In Australia, although marginally improving over the past few years, the fraction of females enrolling in CS/IT education is in the similar range as in the other parts of the world [1]. The lack of gender balance is typically attributed to stereotypes and implicit biases [48], cultural expectations and the lack of role models [2, 40, 53] and lack of early exposure [32, 40] and the educational environment and curriculum [40, 41].

Limited studies have sought the views of higher education educators as to key factors producing this disparity. Our focus in this study was thus to delve into the causes and impacts of gender inclusivity in a CSHE setting by examining the perspectives of CS educators on gender inclusivity within their classrooms. An essential aspect of this investigation is to uncover methods that these educators utilise to deliver an inclusive learning experience. We also wanted to identify any key teaching resources they employ to develop teaching, learning, and assessment content that is more accessible and equitable for all students. We aimed to explore the various factors that influence the participation of CS students in learning activities as perceived by their educators. Such insights are crucial in understanding the dynamics of student engagement and the potential barriers that different genders might face in the educational process. While there has been considerable research on the prevalence of gender bias in CS education and computing in general [13, 19, 26, 31, 40], there remains a scarcity of studies focusing on how educators perceive and address these issues. Our study intends to fill this gap by providing a comprehensive analysis of the strategies implemented by educators to promote an inclusive learning environment.

We present the results of an interview study we performed with 20 higher education professionals in computer science from four universities from *Anon_Region* region (anonymised due to double anonymous requirements). The interviews focused on educators' experiences and views on inclusivity (particularly gender inclusivity) in CS education. Specifically, our study aimed to answer the following three RQs:

RQ1: What is the perception of educators on gender inclusivity in a CSHE environment?

RQ2: Which factors influence gender inclusivity in a CSHE learning environment?

RQ3: How do various student facets, characteristics and behaviours, as perceived by educators, influence participation and engagement in CSHE learning activities?

Our results indicate that educators consistently recognised the gender imbalance in CS classrooms, attributing it to both societal stereotypes and educational practices, emphasising the need for early interventions, gender-neutral language, and diverse representations. They identified several key factors affecting gender inclusivity: early educational experiences and societal norms significantly shape gender participation in CS; inclusive teaching practices, equitable group work, and supportive student-educator interactions are vital for encouraging women students; and institutional policies on hiring and support systems for women faculty and students are crucial. We use *inclusive teaching* to refer to deliberate instructional choices that (i) establish respectful participation norms, (ii) reduce identity-based barriers to engagement (e.g., through language, examples, and interaction structures), and (iii) provide equitable opportunities for students of all genders to contribute and learn. Additionally, the diverse cultural backgrounds of international students impact their engagement, necessitating tailored support. Women students often exhibit strong organisational skills and leadership qualities but prefer private forms of engagement, while male students are more vocal but may resist working with women peers and exhibit biases that require educators' intervention to foster a supportive and inclusive classroom environment. Having said that, we note that while gender disparities in computing and CSHE have been reported across many contexts, the magnitude and drivers of these disparities vary substantially by country and region. Prior

work suggests that the most pronounced and persistent participation gaps in CSHE are often documented in Global North contexts, whereas patterns in parts of the Global South can differ and, in some settings, may be less pronounced or shaped by different sociocultural and economic factors. Accordingly, we use “global” to denote that gender disparity is a widely discussed phenomenon in computing education, while recognising that it is not uniform across contexts and should not be interpreted as equally distributed worldwide.

The key contributions of this research includes:

- Our study presents the perspectives of CS educators on gender inclusivity in CSHE, and key factors influencing gender inclusivity in CSHE.
- Our study further investigates the educators perspectives on student facets and behaviours that influence their participation and engagement in CSHE learning activities.
- We propose recommendations for CS educators and institutes to implement inclusive teaching practices and create supportive classroom environments, and provide insights for designing targeted interventions.
- We also suggest directions for future research, including intersectional analysis, longitudinal studies, and systematic investigation of teaching practices.

In this paper, we recognise that gender identity is a complex and multifaceted construct that extends beyond the binary categories of “men” and “women”. While we try to remain consistent with our terminology, we present the quotes from the educators as-is, without any changes in their quotes. We also acknowledge that our study primarily references “female” (“women”) and “non-binary” individuals in the context of CSHE, and the existence and importance of other gender identities that might not have been explicitly mentioned. We are also aware that the terms “male” and “female” can imply strictly biological distinctions, which are themselves not binary or singular constructs. However, our references to gender are based on self-identification or the perspectives of our interviewees.

2 BACKGROUND AND RELATED WORK

In this section, we synthesise prior work that motivates our focus on educators in CSHE, and we make explicit the theoretical lenses that inform how we interpret educators’ accounts of gender inclusivity.

2.1 Gender disparity in CS and CSHE

Despite decades of attention, women and other marginalised genders remain underrepresented in many computing higher-education contexts. Longitudinal analyses document persistent participation gaps and show that they are maintained by a combination of educational structures, cultural narratives about computing, and institutional pathways that shape entry and persistence. For example, in the United States the share of women in computer science has fluctuated across decades rather than steadily improving, and these patterns are often discussed in relation to shifting cultural stereotypes about computing and the kinds of students who are perceived to “fit” the field [48]. Seminal work in computing education has similarly emphasised that barriers are not only academic preparation or ability, but also the cultural construction of computing as a masculine domain and the exclusionary nature of many learning environments [13, 40].

Importantly, gender participation patterns are not uniform across countries and regions. Cross-national work suggests that participation can vary substantially and may be shaped by the interaction of economic opportunity structures, cultural expectations, and gendered norms about fields of study. Some analyses have reported that more gender-equal countries can exhibit larger gender gaps in certain STEM fields, illustrating that participation is not explained by a single global

mechanism [55]. For CSHE research, this implies that claims about “global” disparity require careful framing: while underrepresentation is widespread, the magnitude, direction, and drivers of gender gaps can differ across sociocultural contexts. This is especially relevant for interpreting educators’ accounts when student cohorts (and educators themselves) may have experiences rooted in diverse cultural contexts.

2.2 Theoretical lenses for understanding participation and persistence

We draw on three complementary lenses that are frequently used to explain differential participation in STEM fields and that map directly onto constructs that appear in educators’ accounts: (1) sociocultural stereotypes and ambient belonging; (2) expectancy-value and social-cognitive accounts of academic/career choice; and (3) identity and intersectionality.

Stereotypes, field culture, and ambient belonging. A robust body of work argues that gender disparities in fields such as computer science are reinforced by culturally shared stereotypes about what computing is and who a “computer scientist” is. These stereotypes can operate as gatekeepers: they shape interest and willingness to participate by signaling whether the field aligns with a student’s values and desired identity [9]. Related work on “ambient belonging” shows that even subtle environmental cues (e.g., stereotypical objects, messaging, or norms) can differentially affect women’s sense of belonging and interest in computer science [10]. At a broader level, reviews of gender gaps across STEM fields emphasize that field-specific cultures and stereotypes help explain why some STEM domains are more gender balanced than others [11].

Within educational settings, stereotype-based mechanisms can manifest through classroom climate, peer interaction patterns, and how competence is recognized. The concept of *stereotype threat* further explains how performance and engagement can be undermined when individuals feel at risk of confirming a negative stereotype about their group, particularly in evaluative contexts [54]. While stereotype threat was developed in a general social-psychological context, it has direct implications for CSHE: assessment-heavy, competitive environments can unintentionally amplify threat, whereas inclusive norms and supportive instructional practices can reduce it.

Expectancy-value and social-cognitive accounts of choice. A second lens focuses on how learners decide whether to enter, persist, and invest effort in a domain. Expectancy-value theory argues that academic choices are shaped by (a) expectations of success and (b) the subjective value of the activity (e.g., interest, utility, attainment value, and perceived cost) [20]. Social Cognitive Career Theory (SCCT) similarly highlights self-efficacy, outcome expectations, and goals as proximal predictors of interest development and persistence, while also emphasizing contextual supports and barriers [36]. Both frameworks are useful in CSHE because educators’ practices can influence students’ expectancies (e.g., feedback, scaffolding, and how competence is framed), subjective task values (e.g., relevance of examples, role models, and career narratives), and perceived barriers (e.g., peer climate, bias, and access to support). These constructs have direct, empirically documented analogues in computing education specifically, not only in STEM broadly. Gender differences in programming self-efficacy have been shown to interact with motivation and goal orientation to predict performance in introductory CS courses, with women students entering with systematically lower self-efficacy despite comparable ability [38], which is consistent with an expectancy-value account in which lower perceived likelihood of success can suppress engagement independent of actual competence. Similarly, cross-national evidence shows that gender gaps in STEM self-efficacy, interest, and aspirations are mediated by cultural and gender norms rather than being fixed or purely individual, and that self-efficacy operates as a key mechanism linking norms to downstream choices [7]. We return to what this mechanism implies for our own findings in Section 7.

Identity, belonging, and intersectionality. A third lens emphasizes that participation is tied to identity: whether students can see themselves as legitimate members of a field and whether the environment affirms or marginalizes salient identities. Work on science and STEM identity (including identity among women of color) shows that recognition, competence, and performance are socially mediated and can vary by context and intersectional positioning [6]. In CSHE, this implies that gender inclusivity cannot be understood in isolation from other identity dimensions (e.g., race/ethnicity, nationality, language background, disability, and socioeconomic status). An intersectional approach is particularly relevant when student cohorts are culturally diverse or include international students, because norms of participation and communication can differ across educational cultures.

Intersectionality as an analytic lens originates in Black feminist legal scholarship, which argued that single-axis analyses of discrimination (e.g., race *or* gender in isolation) fail to capture the distinct, compounded experiences of people who sit at the intersection of multiple marginalized identities, such as Black women [14]. Rather than treating identity categories as additive, intersectionality asks how systems of power (e.g., gender, race, class, nationality) co-constitute one another within a given context, producing experiences that are qualitatively different from – not merely a sum of – the experiences associated with any single identity dimension. Within computing specifically, Rankin and Thomas have argued that HCI and computing research has often applied intersectionality superficially, e.g., as a synonym for ‘diversity’ or as an additional demographic variable, rather than as a structural analysis of how overlapping systems of oppression shape who gets to participate and how [45]. Erete, Rankin, and Thomas extend this critique methodologically, proposing that HCI and design research move away from a presumed position of neutrality and instead explicitly interrogate how structural oppression and power operate across intersecting identities in the settings being studied [22]. We take this as a methodological caution as much as a theoretical one: it is not enough to acknowledge that gender intersects with other identities in CSHE in principle; the analysis itself needs to attend to how those intersections surface in participants’ accounts, which is the approach we adopt when interpreting educators’ narratives about their own identities and about international and culturally diverse student cohorts (Sections 4 and 6).

2.3 Educators as key agents in CSHE inclusivity

Much of the gender-in-computing literature focuses on students, pipeline stages, or broad institutional initiatives. However, educators are direct shapers of the day-to-day learning environment: they determine norms for participation, structure group work and assessment, select examples and representations, and respond to bias when it emerges. Prior work has documented faculty perspectives and concerns about the present and future of CS education, but educator-centred accounts of gender inclusivity practices remain comparatively less common than student-centred studies [39]. This matters because educators’ beliefs about the causes of gender disparity and the “right” remedies can influence which interventions they adopt (e.g., representation-focused materials, inclusive language, mentoring practices) and which structural issues remain unaddressed (e.g., institutional reward structures, staffing, and support systems).

This motivates our study design: by eliciting educators’ lived experiences and interpretations of gender inclusivity in CSHE, we can better understand (a) how educators explain gender participation patterns through the lenses above, and (b) what actions they perceive as feasible within the constraints of curricula, assessment regimes, and institutional policies. In the remainder of the paper, we use these lenses to interpret educators’ accounts and to connect our discussion and recommendations back to established theory and prior evidence.

3 RESEARCH METHODOLOGY

We conducted semi-structured interviews with 20 educators from four universities in *Anon_Region* higher education settings. The four universities were selected to provide variation in CSHE teaching contexts (e.g., cohort composition, unit offerings, and institutional teaching practices) within the same national setting, while remaining feasible for in-depth qualitative inquiry. We recruited the participants via our professional networks and sent out emails with an explanatory statement of our research project (approved by the Human Research Ethics Committee from the university the authors are affiliated to¹.) Participation was voluntary and based on self-selection by responding to the email invitation. We acknowledge that this recruitment approach may introduce self-selection and network-related sampling bias, which we reflect on in Section 8. Our inclusion criteria focused on participants that (i) currently teach, or recently taught, a computing-related unit in higher education, and (ii) had relevant experience teaching in CSHE contexts for at least one year.

In this study, we aimed to answer three key research questions:

- RQ1: What is the perception of educators on gender inclusivity in a Computer Science higher education environment?** This RQ aimed to investigate educators' perceptions regarding gender inclusivity in CSHE environments. With increasing interest in CS, there is a growing need to ensure that the field is inclusive and accessible to individuals of all genders. However, research has shown that CS education environments are still very male-dominated and that women and other marginalised genders may face barriers to participation in the field [41, 61]. As educators play a critical role in shaping the learning environment and culture in the CS classroom, understanding their perceptions of gender inclusivity is essential first step towards gender inclusive CS education environment [59].
- RQ2: Which factors influence gender inclusivity in a Computer Science higher education learning environment?** This RQ aimed to identify key factors that impact gender inclusivity in CSHE environments. We investigate multifaceted aspects that influence gender inclusivity, encompassing educator dynamics, student interactions, course content, and institutional policies, among others. Previous studies have indicated the presence of gender biases and stereotypes in the CS education sector, which can hinder the participation and success of women and other marginalised genders [61]. RQ2 aimed to elucidate the underlying dynamics within CS education environments by investigating these factors. This understanding is crucial for developing targeted strategies and interventions that can foster a more inclusive and equitable educational experience for all genders.
- RQ3: How do various student facets, characteristics and behaviours, as perceived by educators, influence participation and engagement in CSHE learning activities?** This RQ aimed to explore different facets of CS students that influence their participation in learning activities, as perceived by their educators. Understanding these facets is crucial for creating an inclusive and effective educational environment.

3.1 Data Collection

We collected our data using semi-structured interviews to elicit rich accounts of educators' lived experiences and practices regarding gender inclusivity in CSHE. The semi-structured interviews were conducted over six months. Each interview lasted approximately 60 minutes and was recorded with the participants' consent. The interviews explored educators' views and experiences regarding gender inclusivity in CS education, their teaching methodologies, and personal reflections on their professional journeys in CS education.

¹Anonymised

Some interviews (n=7) were conducted in person, and the remaining interviews (n=13) were conducted via the Zoom video conferencing platform due to limited availability, proximity, or interviewees' preferences. At the beginning of each interview, we confirmed that the interviewee had reviewed the explanatory statement and the consent form, which were emailed to them at least 48 hours before the interview². The consent form focused on obtaining consent for the interview recording and on the fact that the interviewees' views could be used anonymously for the publication of our research. We began each interview after confirming that the interviewee had signed the consent form. Each in-person interview was recorded using a dedicated audio recorder, and each Zoom interview was recorded using Zoom's built-in recording functionality.

We began the interviews with general questions (given our main focus on people's lived experiences) on the (i) demographics, e.g., gender, and (ii) experience of teaching in higher education settings, e.g., the courses taught in the past and currently, research areas, and their role at the university³. As part of this initial background, we also asked whether participants had undertaken any special training (e.g., gender-inclusivity or equity-related). Participants did not report any dedicated training in this area, beyond general professional development courses. We also asked about their experiences developing and teaching different subjects and the roles they played in teaching them, e.g., whether they were the convener (responsible for the overall management of a subject) or a lecturer (responsible for delivering the subject content). Thereafter, we transitioned to the main topics we intended to cover in our interviews, namely, educators' experiences working with diverse cohorts (with a focus on gender diversity), the challenges associated with managing such cohorts, attempts (if any) to address these challenges, and examples of such instances. We then focused on educators' teaching approaches, i.e., whether they adapted their instruction to accommodate a gender-diverse classroom or used specific resources for this purpose. We also asked educators about any instances they could recall of a non-inclusive environment and the reasons for such non-inclusivity. In the end, we asked the interviewees whether they would like to provide any further comments or share any additional information. We concluded the interviews by acknowledging their contribution and thanking them for their time.

3.2 Participants

We recruited 20 CS educators for our interviews through professional networks and social media, as summarised in Table 1. We use a pseudonym in the following sections to refer to participants. Ten (10) identified as men, nine (9) as women and one (1) as non-binary. Teaching experiences ranged from 4 to 40 years. The subjects taught by the interviewees varied greatly, however, specialisations included Software Engineering, Hardware, Programming, Cybersecurity, HCI, Information Systems and Cloud Computing. We note that recruitment through professional networks and voluntary participation may have introduced self-selection and network-related sampling bias (e.g., educators with a stronger interest in gender inclusivity may be more likely to participate, and professional connections may shape who is reached). We acknowledge this as a limitation.

3.3 Data Analysis

After each interview, we manually transcribed the recording. For Zoom interviews, we used the transcriptions as the initial version and then revised them by manually reviewing and correcting inconsistencies. While manual transcription is time-consuming [60], we opted not to use any external or automated solution due to privacy concerns and the fact that most of these systems are error-prone, particularly when dealing with diverse interviewee backgrounds. All of our interviews

²Link to Repository with Explanatory Statement

³Link to Repository with the Interview Questions

Table 1. Interview Participant Demographics

Code	Pseudonym	Gender	Teaching Exp. (Y = Years)	Specialisation	Type of Interview
P1	Bobby	Man	4Y	Software Engineering	In-Person
P2	Meera	Woman	20Y	Programming and Algorithms	In-Person
P3	Sara	Woman	15Y	Data Visualisation and Human Computer Interaction	Zoom
P4	Nicky	Woman	15Y	Software Engineering	In-Person
P5	Tai	Woman	3Y	Computer Gaming	In-Person
P6	Alex	Non-binary	10Y	Computer Gaming and Programming	In-Person
P7	Max	Man	20Y	Software and Systems Engineering	Zoom
P8	Tanya	Woman	15Y	Human-Computer Interaction and Robotics	Zoom
P9	Sam	Man	4Y	Software Engineering	In-Person
P10	Dev	Man	12Y	Software Engineering	Zoom
P11	Ash	Woman	10Y	Internet of Things	Zoom
P12	Robin	Man	7Y	Software Engineering	Zoom
P13	Andy	Man	40Y	Data Visualisation and Software Security	Zoom
P14	Leo	Man	21Y	Cloud Computing and Software Systems	Zoom
P15	Ella	Woman	11Y	Optimisation and Software Systems	Zoom
P16	Miles	Man	7Y	Cyber Security	In-person
P17	Evan	Man	4Y	Cyber Security	Zoom
P18	Ivy	Woman	12Y	Information Systems	Zoom
P19	Luke	Man	7Y	Software Engineering and Programming	Zoom
P20	Hans	Man	14Y	Cloud Computing and Internet of Things	Zoom

were transcribed into Microsoft Word documents. All the interviews were analysed and coded in NVivo version 14 - a qualitative data analysis tool. We used NVivo because it supports systematic, auditable management of a large code set (250+ codes, see below) across 20 lengthy interview transcripts, including code hierarchies, query-based retrieval of coded excerpts, and version tracking of code merges and renames, which would have been impractical to manage reliably with manual methods alone [62].

Our data analysis employed thematic analysis across multiple iterations [12]. Specifically, we followed the widely used six-phase reflexive thematic analysis framework of Braun and Clarke [4]: (1) familiarising ourselves with the data, (2) generating initial codes, (3) searching for candidate themes, (4) reviewing themes, (5) defining and naming themes, and (6) producing the final analysis reported in Sections 4–6. Our approach to coding itself (i.e., how individual excerpts were segmented and labelled) drew additionally on coding conventions described by Saldaña [47], including an initial open/first-cycle coding pass followed by a second cycle of pattern coding in which related first-cycle codes were consolidated into broader, more conceptually coherent codes (see the code-merging examples below). The thematic analysis is adopted from the Interpretive Phenomenological Analysis (IPA) [51, 52] and the Grounded Theory approach. Following Smith and Nizza [51], IPA can be used as a complementary approach alongside grounded theory or other theory-building studies: we used IPA to keep our coding oriented towards how educators made sense of and interpreted specific events in their own teaching practice (rather than only their reported behaviours), while using grounded-theory-style constant comparison to build codes and themes iteratively from the data. Although we did not collect or report participants’ individual-level identity or demographic background beyond gender (Section 8), this IPA-informed orientation towards educators’ “lived experience” refers to how they interpreted and narrated specific episodes from their teaching (i.e., a phenomenological stance on the interview data), not to a layered profile of participants’ personal

or cultural backgrounds. Our coding was primarily inductive and data-driven, in that codes and themes were generated from what educators actually said rather than being fixed in advance. However, consistent with reflexive thematic analysis' recognition that analysts are never fully theory-neutral [4], the three theoretical lenses introduced in Section 2 (stereotypes and ambient belonging, expectancy-value/social-cognitive accounts, and identity/intersectionality) functioned as sensitising concepts during theme review and naming: once candidate themes emerged inductively, we used these lenses to check whether a theme captured a construct already well-established in the literature (helping us name and situate it, e.g., linking 'Awareness and Acknowledgement of Gender Disparity' to ambient belonging and stereotype-threat mechanisms) or pointed to something the existing lenses did not fully account for (which shaped, for example, our attention to intersectional patterns in Sections 4 and 6). The lenses therefore informed interpretation and naming of already-emergent themes rather than being imposed as a pre-defined coding frame. The steps followed for our thematic analysis are elaborated in the following subsections. The first author conducted the primary coding of the full dataset. The second author independently reviewed the initial code set and coded excerpts as described below, and the third author participated in the review and refinement of themes. Throughout the analysis, we maintained an audit trail (e.g., code merges/renames and theme revisions) to ensure traceability from excerpts to codes and themes.

Familiarization with Data: After collecting the data, the first author immersed themselves in the data, reading and re-reading the content, noting initial ideas and patterns.

Generating Initial Codes: The next step involved generating codes from reviewing the interviews to identify important features of the data relevant to our research questions. This was an open coding phase in which data were segmented without attempting to fit them into a preexisting coding frame. This first-cycle, open-coding pass followed the coding conventions described by Saldaña [47], in which excerpts are labelled with short, descriptive 'first-level' codes close to the language and meaning of the data, deferring more conceptual, cross-cutting groupings to the later pattern-coding and theme-generation phases described below. This involved a detailed examination of educators' responses concerning their teaching styles, experiences, and approaches to inclusivity. We note that we did this process in two iterations. In the first iteration, we coded six interviews from our interviewees and generated 490 codes across all three RQs, grouped into categories such as educators' perceptions, students' behaviour, teaching methods, gender bias, and gender-inclusive practices. To ensure the reliability and accuracy of the thematic coding, the codes were independently reviewed and validated by the second author. We reviewed each code and identified any duplicates or discrepancies. The duplicate removal process led to 399 codes. Thereafter, we reviewed all 399 codes and the corresponding coded interview content. We merged, renamed, and recoded the data to ensure consistency across all interviews, based on discussions between the first and second authors. This merging step is what Saldaña [47] describes as second-cycle 'pattern coding': related first-level codes are consolidated into broader, more conceptually coherent codes that better capture a shared underlying idea. For instance, the first-level code, "Application of Inclusivity in Course Content Development," was merged into the "Adapting Teaching Approaches Based on Subject Matter" code. Another example codes "Fostering Critical Thinking in Students", "Fostering Holistic Development in Students" were merged into the code "emphasising the Importance of Preparing Students for Diverse Workplaces and Real Life Scenarios". This process yielded 252 codes in total, which were consistently applied across all six interviews. Thereafter, we completed coding the remaining 14 interviews, using these 252 codes as the basis. We continued to add new codes; however, we sought to maintain consistency across all previously coded interviews. The final set comprised 282 codes. We note that we did not compute a numerical inter-rater reliability (IRR) statistic because our coding involved iterative refinement of the codebook and code definitions

through discussion; instead, we focused on consistency through independent review of the initial codes, removal/merging of duplicates, and consensus-based refinement of code definitions that were then applied across the full dataset.

Searching for Themes: Codes were then collated into potential themes. This involved analysing the codes and collating data to identify significant broader patterns of meaning (potential themes). We sought patterns and recurring themes in how educators described their classroom dynamics, challenges, student interactions, and their educational philosophies. Key areas, including their awareness of gender issues, methods for addressing gender balance in classrooms, and their personal teaching experiences, were analysed. Each instance in which an educator mentioned practices or beliefs that could influence gender inclusivity was included in the thematic analysis. Overall, we identified 10 themes for RQ1 and RQ2, and 6 for RQ3. Concretely, this step involved exporting all coded excerpts per code from NVivo, clustering codes that referred to a common underlying concept (e.g., separate codes describing different classroom strategies for encouraging quiet students were clustered under a broader ‘inclusive teaching practices’ candidate theme), and re-reading the underlying excerpts for each cluster to check that a single candidate theme could coherently describe them. Where the codes within a cluster were about the same surface topic but described conceptually different mechanisms (e.g., practices aimed at students versus practices aimed at institutional hiring), we split them into separate candidate themes at this stage rather than in the later reviewing phase.

Reviewing Themes: After having identified themes, they were then reviewed and refined in discussion with the second and the third authors. This involved checking the themes against the dataset to determine if they form a coherent pattern. At this stage, several themes were split, combined, or discarded. The final set of themes were six themes for RQ1, four main themes for RQ2, and three main themes for RQ3.

Defining and Naming Themes: Clear definitions and names for each theme were then developed, which involved refining the specifics of each theme and the overall story for our analysis. The resulting themes are discussed in the next sections for each RQ.

4 RQ1 – PERCEPTION OF EDUCATORS ON GENDER INCLUSIVITY IN CSHE SETTINGS

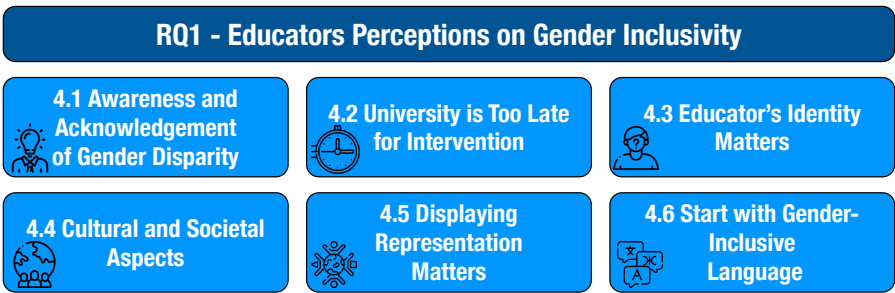


Fig. 1. RQ1 Key Themes

In RQ1, we aimed to gain insights into a range of educators’ perceptions of gender inclusivity in their CSHE environment. We sought to understand educators’ awareness and perspectives on gender disparities and inclusivity within their classrooms and institutions. We aim to uncover

Table 2. Mapping: Educators and RQ1 Themes Mentioned

Educator	Awareness (1)	Uni Too Late (2)	Educator's Identity (3)	Cultural & Social Aspects (4)	Displaying Representation (5)	Gender-Inclusive Language (6)
Bobby [P1]	✓	✓		✓		✓
Meera [P2]	✓		✓		✓	✓
Sara [P3]	✓		✓		✓	✓
Nicky [P4]	✓	✓	✓	✓	✓	✓
Tai [P5]	✓	✓	✓	✓	✓	
Alex [P6]	✓	✓	✓	✓	✓	✓
Max [P7]	✓	✓		✓	✓	
Tanya [P8]	✓	✓	✓	✓		
Sam [P9]	✓	✓	✓	✓		
Dev [P10]	✓	✓		✓	✓	
Ash [P11]	✓	✓	✓	✓		
Robin [P12]	✓	✓		✓	✓	
Andy [P13]	✓	✓		✓	✓	✓
Leo [P14]	✓	✓		✓	✓	
Ella [P15]	✓	✓	✓	✓	✓	✓
Miles [P16]	✓			✓	✓	
Evan [P17]	✓	✓		✓	✓	
Ivy [P18]	✓	✓	✓	✓	✓	
Luke [P19]	✓	✓		✓	✓	✓
Hans [P20]	✓	✓		✓	✓	
Total	20/20	17/20	10/20	18/20	16/20	8/20

educators' key insights into the existing state of gender inclusivity, their observations of gender-related challenges, and the potential barriers women and other marginalised genders face in the CSHE field. The emphasis here is gathering qualitative insights into the educators' understanding and attitudes towards gender inclusivity. RQ1 serves as a basis for exploring more in-depth factors in subsequent questions. Fig. 1 outlines the key themes we found on analysing our interviews, and Table 2 maps the themes to the educators who discussed these themes in their interviews. To support interpretability, we also report the number of participants whose excerpts contributed to each theme (from the overall $n = 20$ participants). These counts are intended as descriptive context to indicate how widespread a theme was within our sample; they should not be interpreted as statistical estimates of prevalence beyond the studied context.

4.1 Awareness and Acknowledgement of Gender Disparity

Our interviewees demonstrated varying levels of awareness regarding the gender gap in CSHE, acknowledging differences in enrollment and participation among different genders in their classes and experience. However, all 20 educators (irrespective of their background) noted the gender imbalance in their CS classrooms, with a significantly smaller proportion of women students in their classrooms, or in other words, acknowledged a male-dominated classroom environment. Most educators noted that the ratio is generally at 70:30 at best in most CSHE undergraduate or postgraduate classrooms:

“In general... there's always gender imbalance when you go to any IT classroom, so you will never see 50-50 balance between the cohort. (...) It's probably not good but I'm always happy there is at least 25% female students in the classroom. - Bobby [P1]”

“It[Class Distribution] is pretty tough in terms of gender diversity. I mean, it is skewed, obviously, mostly males. I mean, that’s fairly obvious. - Sara [P3]”

Educators like Alex and Max further confirmed the gender disparity, acknowledging the skewed gender balance in certain units, particularly those perceived as more technical or traditionally male-dominated. Alex’s observation of “*next to no women*” in some classes underscores the significant gender gap in enrollment. Miles and Evan, focusing on cybersecurity, highlighted a particularly poor women participation rate (compared to other CS courses), underscoring the need for targeted initiatives in certain sub-fields of computer science. Ivy also observed stark contrasts in gender ratios depending on the technicality of the unit. Some educators with backgrounds in disciplines other than CS highlighted the stark contrast in gender ratios they have experienced, underscoring the uniqueness of the CS gender gap. Sara [P3] from a Psychology background noted:

“I went from ‘Oh, my God, there’s a boy in the class [in psychology], how exciting’ to ‘Oh, my God, there’s a girl in the class. Oh, how exciting.’ So it was a real swap from the gender ratios from psychology into IT. - Sara [P3]”

This awareness of gender disparity among educators extended beyond mere acknowledgment – it revealed a resigned acceptance of many to this imbalance as an established norm within the CSHE landscape. Similar to Sara’s (P3) thoughts, while all educators noted the lack of female students in CS undergraduate and postgraduate education, they also seemed to believe that this is ‘fairly obvious’. This tacit acceptance underscores a broader, more systemic issue: the normalization of gender imbalance in STEM fields, particularly in CS [40, 48]. The educators’ comments reflect a recognition of the problem and also hint at an underlying sense of inevitability regarding the gender dynamics in their classrooms.

“I think I’m very used with that [now]. And I feel it’s [a] normal thing. - Tanya [P8]”

Several educators also noted that progress has been made regarding gender diversity over the past few years. They noted an increased acceptance and normalisation of non-binary and transgender students in the student cohorts, which was not the case earlier in their teaching careers. Some educators also mentioned that their perceptions of gender ratios of men, women or non-binary are sometimes only based on appearances, and hence demonstrated a nuanced level of awareness of gender issues. For instance, Luke (P19) repeatedly mentioned that his observations on gender in classrooms are based on “female-appearing”, “female-looking” or “cis-female” students. This reflects a broader societal shift towards inclusivity and acceptance.

“I think people are a lot more accepting, and even expecting of differences in how the student cohort looks – non binary students, trans students, I mean, that wasn’t something we thought about when I was a student. - Meera [P2]”

This collective insight and acknowledgement from educators highlights a critical issue in CSHE: the persistent gender disparity that hinders the achievement of a more diverse and inclusive academic environment. Despite variations in experiences and the degree of disparity across different institutions and units, the consensus among educators underscores the need for coordinated efforts to address and mitigate gender imbalances [24].

4.2 University is too late for any intervention for gender imbalance

Some of our educators believe that university interventions might be too late to significantly impact the gender imbalance in CS, highlighting the crucial role of early education in shaping perceptions and aspirations towards the field. This underscores a broader societal and educational

challenge, suggesting that foundational attitudes and stereotypes surrounding CS education (or broader STEM subjects) are established well before students reach tertiary education [1, 32].

“*I guess part of it [decision making] is also what happens in the out of primary school and high school. There is this discussion all around math, and these things are hard [for female students], and digital (...) computer science, or we have the science word in it (...) makes it a concern for some people. And typically, this goes more unfortunately to the female, not on the boys side, which is very frustrating.* - Max [P7] **”**

Educators argue that the societal and pedagogical framing of CS education as challenging or not suited for certain genders deters young girls from pursuing these subjects. This early discouragement is further exacerbated by a lack of gender-inclusive representation and encouragement within educational settings, leading to a diminished female presence in CS courses at the university level.

“*I think school is very important because that's where you actually, you kind of get to discover skills. And if when you discover your skill, someone said, 'Oh, by the way, this is not for you.' And everyone around you saying the same.* - Max [P7] **”**

The overall consensus among interviewees is that interventions to promote gender inclusivity in CS need to start much earlier than university. Efforts should focus on primary and secondary education levels, where the foundation for future educational and career choices is laid [1, 32].

4.3 Educator's Own Identity Matters

The identity and personal experiences of educators play a pivotal role in shaping the dynamics of gender inclusivity within CS classrooms. We note that all educators who identified as females or non-binary (except Luke (P19)) shared how their identity plays a pivotal role in shaping their perceptions and strategies for fostering an inclusive environment. Luke mentioned that he was raised in a country where females have a much more empowered role in society (than in some other parts of the world), which influences his identity as a person and CS educator.

Educators' own experiences or identities as non-male individuals in the field of computer science enable comfort for students facing challenges. They demonstrate that shared gender identity or experience with marginalized groups can foster empathy and a supportive teaching approach. Tanya (P8) also shared an incident where a female student felt uncomfortable in a classroom full of male students. The representative quotes below underscore the vital role of educators' identities in developing empathetic teaching practices that address the needs of diverse student cohorts.

“*I have had multiple situations where they [female PhD students] will come specifically to me [to discuss any issues], because they feel more at ease (...) they would never go to their male supervisors?* - Nicky [P4] **”**

“*Being openly non-binary (...) I think does make some students, particularly some of the queer students feel more comfortable to come and speak to me [about any issues in the classroom]* - Alex [P6] **”**

Educators' transparency about their identities can foster an inclusive classroom environment [43]. By openly discussing their identity, Alex creates a space where diversity is celebrated and students feel comfortable expressing themselves, particularly benefiting those from queer communities. Similarly, another female educator shared that being openly Hijab-wearing creates an inclusive classroom environment. This aspect of educators' identity matters significantly, as it directly impacts the inclusivity of the learning environment and supports students in feeling seen and valued.

Beyond fostering inclusivity, female (non-male) educators also discussed their unique challenges that underscore broader systemic issues within the academic and CS communities. Some educators reflected on the difficulty of being a female (non-male) educator in the CSHE environment. For instance, Tai [P5] discussed her experience:

“People have seen my name on the teaching list, and [assume] it’s going to be some guy. And then they go to class, and they’re very confused when they see me and [ask] where’s the teacher?” - Tai [P5]

Tai’s experiences show the stark reality of gender assumptions in CSHE. Her narrative reveals a notion of the expected gender identity of educators, leading to confusion and questioning of authority. This confusion comes across as not merely a case of mistaken identity but a symptom of deeper societal and cultural expectations about gender roles within CSHE (which we discuss in detail later in this section). Tai and other educators also alluded to the perceived lack of authority in female educators’ classrooms compared to their male counterparts, and the issue of perceived authority extends beyond initial assumptions about gender. Tai’s observation that female educators are often seen as “easy targets” compared to their male counterparts, touches on gender dynamics in power and authority within CSHE settings. Ash shared that as a female educator in a fairly technical field of Internet of Things or embedded systems, she must constantly prove herself to students and male counterparts.

“I think that they do not want to confront a male tutor, right? They feel like I’m an easy target. Whereas with a male tutor they always feel like this person (...) has more authority and has more power. (...) I guess they don’t see that in me. - Tai [P5]

“I have to prove myself. I have to do an extra bit of work to show my male counterparts that I can do it, too. (...) I and another colleague were doing the tutoring, which was all hands-on and hardware. Every time there was a question they (students) asked him and would not ask me. So it took them like 2 or 3 weeks to realize (that I can answer questions too). - Ash [P11]

Educators like Tanya (P8), Nicky (P4), and Alex (P6) leverage their identities to practice empathy, understanding, and inclusivity. However, they also discussed the systemic challenges faced by women and non-binary educators, such as battling stereotypes and asserting their authority, as shared by Tai (P5) and Ash (P11). From the male educators, Luke and Evan stood out with their perceptions and passion towards ensuring an inclusive, empathetic and approachable classroom environment for women students, as they both alluded to the biases and ‘hardships’ in the journey of women students to be in the CSHE. Overall, the experiences of all these educators show the necessity for ongoing discussions and actions towards dismantling gender biases and fostering an equitable academic environment in CSHE.

Read through the intersectionality lens introduced in Section 2, these narratives are notable for how gender identity alone does not fully account for the dynamics educators describe – gender consistently intersects with a second, co-present identity dimension to shape a distinct experience. Alex’s account of being approachable to ‘particularly some of the queer students’ is not simply about being a non-binary educator in the abstract; it is the co-presence of a non-normative gender identity *and* openness about it that appears to signal safety specifically to queer students, illustrating how gender and sexuality intersect to produce a form of relatability that neither dimension would produce alone [14, 45]. Similarly, the educator who described being openly Hijab-wearing as fostering an inclusive classroom is describing an intersection of gender and religious/cultural visibility, not gender in isolation. And Tai’s and Ash’s accounts of having to ‘prove’ themselves are concentrated in technical sub-fields historically coded as masculine (computer gaming, Internet of Things and embedded hardware), suggesting that the authority penalty women educators describe is

not uniform across CS but is amplified at the intersection of gender and field-level stereotypes about who ‘belongs’ in more technical specialisations [6, 9]. Consistent with calls to treat intersectionality as a structural analytic rather than an additional demographic label [22], we read these as cases where a single-axis account of ‘gender’ in CSHE would miss why particular educators experience inclusion, scrutiny, or authority challenges differently from one another.

4.4 Cultural and Societal Aspects

The cultural and societal influences on gender inclusivity in CSHE reveal complex dynamics that shape student participation and educators’ perceptions. Educators like Evan (P17) and Max (P7) identify a pervasive stereotype that demarcates CS as a “boys’ club” deterring women participation from an early age. Evan points out the unfortunate attachment of cybersecurity and CS to male dominance. Max (P7) shares a personal anecdote of his daughter’s reluctance to engage in a robotics course (due to being the only girl), emphasising girls’ early discouragement due to such stereotyping.

“There is some sort of stereotyping on the computer science and software engineering. This is kind of like a, you know, boys thing and not girls thing... she [Max’s daughter] found that everyone there [in the course] was a boy. And that was the end of this. - Max [P7]”

“This thing called boys club, or things unfortunately attached to computer science or cybersecurity that need to be removed. - Evan [P17]”

Ella (P15) and Leo (P14) discuss how early socialization into gender roles impacts students’ interests and participation in CS. Ella observes girls’ association with traditionally feminine activities and boys with video games and computers, suggesting that societal perceptions of gender roles begin influencing career interests from a young age. Ash shared her own example and mentioned that she was actively discouraged from following a fairly technical field of computing hardware from her student times, as traditionally, that’s “not a female field”.

“For girls (from an early age), it’s bobbies and pink and that sort of thing. But boys, video games and that sort of thing. So, they go on with video games, computers, STEM, and science. - Ella [P15]”

The majority of the educators we interviewed, were not born and raised in ‘Western-world’ countries, and hence they often alluded to the lack of gender disparity concerns in their own context (growing up) and the contrast they witness in the *Anon_Region* context. Interestingly, all educators contributed to their perspectives on cultural and societal aspects, except for the only two women educators born and raised in *Anon_Region*. The educators emphasised that in non-Western countries, such as India, Egypt, China, Philippines, Sri Lanka, and Bangladesh, the gender ratio in the CS education system is much more balanced than in *Anon_Region*.

“I literally lived in a bubble (...) to grow up in a country in the Southeast Asian countries not knowing that cultural problem around the female child [pursuing STEM]. I’m not talking about not experiencing it, but not even knowing it [that any gender barrier in STEM] exists. - Nicky [P4]”

“Internationally, we get a lot of students from the Indian subcontinent... and also a lot of students from China. In these countries, even in computer science, you see much more female participation as compared to what you see in *Anon_Region*. - Bobby [P1]”

Some educators pointed out differing gender imbalance at the undergraduate and postgraduate levels. Statistically speaking, there is a larger proportion of ‘domestic’ students at the undergraduate

level and more international students in postgraduate courses ⁴. Consequently, there is often a better gender ratio at the postgraduate level.

Several educators, such as Tanya, Andy and Sam, discuss how the cultural backgrounds of international students influence classroom dynamics and gender inclusivity. While previous examples focused on the gender dynamics of students, here, educators focused on their perception of students' reactions towards the gender of educators. Tanya notes the comfort level of local students compared to the occasional discomfort among international students to women tutors, while Sam observes differences in student behaviour and engagement based on their cultural upbringing. On similar lines, Andy mentioned:

“ *Students coming in from a predominantly male society. They tend not to ask questions to female tutors, and most likely look for male tutors instead.* - Andy [P13] **”**

This theme reveals that cultural norms and societal stereotypes affect gender inclusivity in CSHE. Understanding these influences is crucial for developing effective strategies to promote gender diversity in the field. By acknowledging and addressing the multifaceted cultural and societal factors, educators and institutions can create more inclusive and equitable learning environments for all students.

4.5 Displaying Representation Matters

More than half of the educators interviewed stressed the importance of having women instructors and tutors as visible student role models. Sara actively strives to employ women tutors to ensure that students see women in authoritative and knowledgeable roles within CSHE, challenging the field's gender stereotypes. Meera echoes this sentiment, emphasising efforts to achieve gender balance within the teaching team to make all students feel represented and welcomed. Max noted the encouraging effect on women students when they see women lecturers, tutors, or peers actively participating in STEM activities. Similarly, Ash mentioned that open days are the perfect avenue to showcase women students' projects, aiming to inspire prospective students by highlighting the achievements of women within their department.

“ *It would be good to see more women [among students]. We try and get that on our teaching team. (...) it's nice to try and get a balance at least reflected there as much as we possibly can. (...) I think that then encourages the students if they see more people like them in the teaching team and that helps them feel that they're in the right place.* - Meera [P2] **”**

Some educators like Alex highlighted the deliberate choice to include diverse case studies in units, aiming to present various perspectives and experiences within the CS field. This effort to incorporate varied narratives and backgrounds into educational content is seen as a way to broaden students' understanding of who can be successful in CS and what that success can look like.

“ *In some of the other weeks when we're dealing with case studies...we do go out of our way to try and get different case studies from different games from different developers that might have different lived experiences* - Alex [P6] **”**

While the primary focus of this study is on gender representation, some educators, such as Alex, Bobby, Nicky and Meera, suggested that representation should also encompass other dimensions of diversity, such as cultural and experiential backgrounds. This broader approach to representation acknowledges the multifaceted identities of students and the importance of creating an educational environment that reflects the diversity of the student body and the wider society.

⁴Anonymised

4.6 Use of Gender Inclusive Language

Several educators like Bobby, Meera, Sara, Nicky, Alex, Andy, Ella and Luke emphasised the need to use gender-neutral language in classrooms and choose examples and activities that do not perpetuate gender stereotypes.

“So it’s a matter of making sure you use a range of genders reflected in the types of names that you use. And you’re not just talking about ‘he’ all the time that you mix it up a little bit, or make it default to ‘she’ (...) I suppose just using inclusive language and inclusivity in the examples of teaching that you’re giving to students [matters]. - Meera [P2]”

Bobby, Nicky, Meera and Ella emphasise the need for (formal) training programs to educate teaching staff on the importance and implementation of gender-neutral language. Such training could include explicit guidelines on using gender-neutral pronouns and ensuring educators are conscious of inclusivity in their teaching methods. They mention that it needs to be a combination of institutional initiatives and individual efforts.

“It’s a matter of university coming up with training program... Like for example, just using gender-neutral pronouns, could be part of that training. - Bobby [P1]”

Educators like Andy, Bobby, Luke and Ella highlight how language use directly influences classroom dynamics and students’ sense of belonging. Andy and Luke’s focus on avoiding male-oriented pronouns or using names to avoid misgendering students shows a conscious effort towards inclusivity. Ella and Bobby shared anecdotes of intervention when encountering non-inclusive language from their tutors, reflecting a proactive stance on creating an environment where all students feel acknowledged and respected.

RQ1 Key Findings:

- **Awareness and Acknowledgement of Gender Disparity:** All 20 educators recognised the gender imbalance prevalent in CS classrooms, reflecting a broad consensus on the existing state of gender inclusivity (or lack thereof) that, for many, had settled into a resigned acceptance of disparity as a persistent, near-universal feature of the field.
- **University is Too Late for Any Intervention:** Many educators felt that university-level interventions come too late to meaningfully shift the gender imbalance, pointing instead to primary and secondary education as the stage where foundational attitudes toward CS and STEM are formed.
- **Educator’s Own Identity Matters:** Educators’ own gender, and other intersecting facets of their identity (e.g., sexuality, religious/cultural visibility, and specialisation area), shaped both the empathy they could offer students and the authority challenges they personally faced.
- **Cultural and Societal Aspects:** Educators, most of whom were themselves educated outside *Anon_Region*, contrasted the pronounced gender gap they observe in *Anon_Region* with the comparatively more balanced gender ratios in CS education in their countries of origin, pointing to cultural and societal norms rather than an inherent property of the discipline.
- **Displaying Representation Matters:** Visible representation of women and non-binary staff, and diverse case studies and role models in course content, was widely seen as important for signalling belonging to students of all genders.

- **Use of Gender Inclusive Language:** Educators emphasised gender-neutral language, and formal training on inclusive language, as practical, low-cost levers for fostering an inclusive classroom.

5 RQ2 – FACTORS THAT INFLUENCE GENDER INCLUSIVITY IN CSHE ENVIRONMENTS

We aimed to identify the key factors that our educators believed influence gender inclusivity in their CSHE learning environments. These factors reported range from educator dynamics, student interactions, and course content to broader institutional policies and societal norms. Based on our interview analysis, the factors influencing gender inclusivity in CSH have been categorised into four main areas: external influences, classroom dynamics, student-educator interaction, and institutional environment. Fig.2 summarises the key factors/themes we found after analysing the interview data, and Table 3 maps these themes against the educator names that discussed these aspects in their interviews. *External influences* include early education and outreach programs promoting computer science to female high school students and societal stereotypes discouraging girls from pursuing STEM fields. *Classroom dynamics* focus on the roles and behaviours of educators, peer interactions among students, and the inclusivity of course content and teaching methods. Within classrooms, a noticeable facet is the *interaction dynamics of students and educators*, which focus on approachability of educators, and the level to which educators make everyone feel welcomed in the classrooms. The *institutional environment* encompasses policies and practices related to hiring female faculty and support systems for female students. By examining these influencing factors, we seek to uncover CS education’s systemic and operational aspects that impact gender inclusivity, providing a foundation for proposing specific interventions.

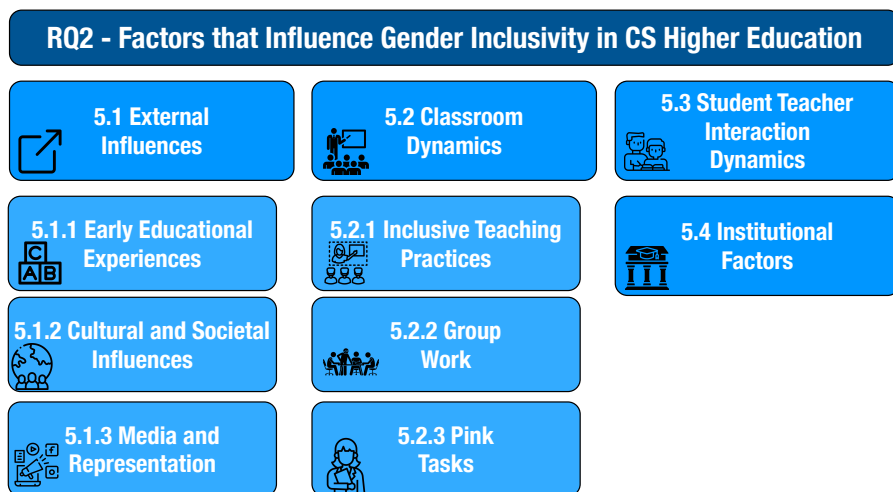


Fig. 2. RQ2 Key Themes

5.1 External Influences

According to our interviewees, external influences play a crucial role in shaping gender inclusivity in CSHE. Early education, i.e., primary and secondary education before entering the university, plays

Table 3. Mapping: Educators and RQ2 Themes Mentioned

Educator	External			Classroom Dynamics			SDE	Instt.
	Ext.T1	Ext.T2	Ext.T3	CD.T1	CD.T2	CD.T3	SDE.T1	Instt.T1
Bobby [P1]	✓	✓		✓	✓			✓
Meera [P2]			✓	✓			✓	✓
Sara [P3]			✓	✓	✓	✓	✓	✓
Nicky [P4]	✓	✓		✓	✓		✓	✓
Tai [P5]	✓	✓	✓	✓	✓	✓	✓	
Alex [P6]	✓	✓	✓	✓	✓	✓	✓	✓
Max [P7]	✓	✓		✓			✓	
Tanya [P8]	✓	✓			✓		✓	
Sam [P9]	✓	✓	✓				✓	
Dev [P10]	✓	✓	✓	✓				✓
Ash [P11]	✓	✓	✓	✓	✓	✓	✓	✓
Robin [P12]	✓	✓	✓	✓	✓	✓		
Andy [P13]	✓	✓		✓				✓
Leo [P14]	✓	✓		✓	✓			
Ella [P15]	✓	✓	✓	✓	✓	✓	✓	✓
Miles [P16]		✓		✓			✓	✓
Evan [P17]	✓	✓	✓	✓		✓	✓	
Ivy [P18]	✓	✓	✓	✓	✓		✓	
Luke [P19]	✓	✓		✓	✓		✓	
Hans [P20]	✓	✓	✓	✓			✓	
Total	17/20	18/20	12/20	18/20	14/20	7/20	15/20	10/20

a pivotal role in shaping perceptions and interests, with stereotypes and biases often discouraging girls from pursuing CS (also covered in Section 4.2). Cultural and societal norms further influence these perceptions, with varying female participation observed across different countries of origin of our interviewees. Additionally, media portrayals and societal stereotypes continue to reinforce traditional gender roles, affecting students' views on CS. Understanding these external factors is essential for developing comprehensive strategies to encourage more women and marginalised genders to pursue and succeed in computer science.

5.1.1 Early educational experiences. Early educational experiences significantly influence students' interest and confidence in STEM fields (n=17), directly impacting their participation in CSH [42]. The quality of math and science education and the encouragement students receive shape their attitudes toward these subjects, which are foundational for pursuing CSHE. Andy [P13] noted a trend of dropped interest of students towards CS-related subjects in secondary school, noting that *“something must have happened between primary school and secondary school”* to cause female students to lose interest in CS-related career choices.

Educators play a critical role in shaping students' attitudes toward STEM. Max [P7] highlights the impact of messaging around STEM subjects like mathematics on students' confidence and interest. He recalls *“When my daughter was in Grade 3, many of her teachers said that maths is very hard, so it's okay if you don't like maths. (...) part of it [the dropout of students] is the message that they get from school.”* This discouragement in early education often deters students, especially girls, from pursuing STEM due to deep-rooted gender stereotypes. Ash [P11] shares a personal experience of being discouraged despite excelling in mathematics: *“So many people discouraged me when I wanted to go down this path with maths. A teacher came and told me don't do it [despite that she was doing well in Maths]. Go and become a doctor. (...) It's (Maths) the male thing. It's not a girl thing.”*

Ivy [P18] echoes this sentiment, emphasising the influence of educators on students' liking for a subject (and excelling in it), particularly when educators build strong relationships with their students. On similar lines, Evan [P17] considers the pivotal role of school career counsellors and awareness programs in guiding students toward STEM careers. He stresses the importance of informed career guidance: *"The career counsellor in schools is a very, very important role. Many of these career counsellors are not even aware of the possibility of career in cybersecurity."*

Encouraging students to explore STEM subjects from an early age can significantly impact their career choices. Exposure to technology and programming in primary and secondary schools can foster interest and help address gender stereotypes.

“It [encouraging female students to try CS] should start from the primary school or secondary school level because we need to encourage them to try different things. - Tanya [P8]”

5.1.2 Cultural and Societal Influences. Societal and cultural influences play a significant role in shaping gender inclusivity in CSHE (n=18). These influences often manifest through cultural attitudes, and societal expectations and norms, which impact female students' general perceptions and decisions regarding STEM fields and consequently impact their uptake of the CS field. As highlighted by the interviewees, general societal perceptions reflect a persistent stereotype that computer science and related fields are predominantly for males. For instance, Max [P7] noted, *"There is some sort of stereotyping on computer science and software engineering as boys' things"*. This cultural bias of considering CSHE as 'geeky stuff' discourages many young women from pursuing careers in these areas from an early age.

Cultural and economic aspects also influence gender inclusivity. In some regions, studying IT is considered prestigious, and furthermore, the economic drive for stable and lucrative careers can either encourage or dissuade females from entering STEM fields. Dev [P10] pointed out that *"job prospects and family pressures"* are significant factors that determine students' career choices. Bobby [P1], on similar lines, mentioned that some countries like India, Bangladesh, Sri Lanka and Iran, see better gender ratios in IT, as it is easier to get a job and livelihood with IT, whereas in countries like *Anon_Region*, studying IT/STEM is not a pre-requisite for getting a 'well-paid' job or a decent standard of living. Our educators interviewed in this study were born in different countries, except three, who were born and raised in *Anon_Region*. All remaining 17 educators mentioned that the gender balance in CSHE is better in their countries, e.g., Albania, Bangladesh, China, India, Iran, the Caribbean, and the Philippines, than in *Anon_Region*, or several other 'western countries'. There are several underlying factors beyond the scope of this study that contribute to this discrepancy, including cultural attitudes towards education, societal norms, e.g., 'boys and girls' playing together as kids, economic incentives, and societal expectations.

“In my country, the people who study STEM, engineering, mathematics, they are seen as smarter... but here in *Anon_Region*, even if they enjoy programming, they still prefer to stay in fields like music, commerce, and so on. - Dev [P10]”

5.1.3 Media and Representation. Media and representation emerged as significant factors influencing gender inclusivity in CSHE (n=12). The portrayal of gender roles in media and the visibility of women in STEM fields play crucial roles in shaping societal perceptions and aspirations. Ella [P15] highlighted how societal norms depicted in media, such as the idea that *"girls are supposed to be with barbies and pink stuff and boys do gaming"* reinforce gender biases from a young age. This type of media representation can deter young girls from developing an interest in CS and related fields. Alex [P6] and Tai [P5] pointed out the shift in media towards more inclusive and diverse representations in recent years. For example, Alex [P6] mentioned *"There's been a shift in the media,*

particularly in games around recent years to be much more inclusive and diverse.”, which helps to create a more inclusive environment and challenge traditional gender norms.

Interviewees also emphasised the need for diverse and positive representation via role models in media to counteract gender stereotypes. Sara [P3] and Alex [P6] noted the importance of showing successful women in technology in various media forms and industry forums to inspire female students and demonstrate that careers in STEM are attainable and rewarding for women. Ash [P11] and Alex [P6] mentioned the impact of having female ambassadors and role models, in media and on events such as open days, stating that seeing women succeed in these fields helps to build confidence and interest among female students and their parents [2].

“ *Start (representation) from things like media, promotional materials, leadership and invited guests. (...) Even a brochure featuring all males can instantly disconnect potential female students. It's essential to convey that no career is inherently boyish or girlish - every career is for everyone.* - Evan [P17] **”**

5.2 Classroom Dynamics

Classroom dynamics play a crucial role in fostering gender inclusivity in STEM fields. Understanding these dynamics is essential for creating an environment where all students feel supported and encouraged. We identified three main sub-themes under this theme: inclusive teaching practices, observations from group work and ‘pink tasks’. To further explore how classroom dynamics influence gender inclusivity, we delve into the specific aspects of teaching practices, group work, and task assignments, as discussed by our interviewees.

5.2.1 Inclusive Teaching Practices. Inclusive teaching practices promote gender inclusivity in CSHE. By fostering an interactive, inclusive, and encouraging classroom environment where all students feel valued and supported, educators can help bridge the gender gap and make students feel inclusive (n=18). Inclusive teaching practices in CSHE focus on encouraging student engagement through interactive and gender-neutral approaches, creating a supportive environment with diverse examples and case studies, and providing constructive feedback to enhance student confidence and participation. Techniques like flipped classrooms, peer reviews, and inclusive language help ensure all students feel seen and valued, fostering a more inclusive and supportive educational experience.

The majority of the educators (n=14) did not mention anything explicitly related to the teaching content influencing gender inclusivity in CSH. Bobby [P1], Nicky [P4] and Dev [P10] mentioned that ‘gender is not a lens’ when developing the course content or what they teach. Evan [P17] noted a major difference in the female cohort size in two cybersecurity specialisations at his university - one focused on technical aspects and the other on the business operational aspects, with the business specialisation with a better female-male ratio, hinting at the impact of changes in course structures on gender inclusivity. Sara [P3], Tai [P5] and Alex [P6] briefly discussed how they are mindful of topics when selecting case studies for their units, e.g., Tai [P5] and Alex [P6] pick case studies or project topics in gaming that reflect diverse perspectives (e.g., queer games) or, in the case of Sara [P3], provide additional support and material for a case study on ‘selecting outfits software’ where she felt that the male students could have a ‘disadvantage’. Nicky [P4] noted that the technical topics in CS do not require any gender lens, as there is little scope for any opinions or values, whereas, in topics such as gaming and human-centred design, there is a scope for students to bring in their perspectives and gender being a central topic. Bobby [P1] mentioned that it is important to maintain gender-neutral language even with very technical topics. He recounted an experience of a co-teacher referring to technical concepts such as ‘hard drive’ and ‘server’ using he/him pronouns, highlighting the need for inclusive language even in technical discussions. Nicky [P4] in a similar vein, recalled an incident from her unit where a project on the game ‘Nine Men’s

Morris' led to a discussion with the students where they felt that the name was not inclusive and changed it.

Implementing practices that ensure all students feel welcome and capable of contributing can significantly impact their engagement and confidence. Bobby [P1] and several other educators highlight the challenges some female students might face in terms of participation and confidence in technical subjects or simply feeling shy, suggesting tailored strategies to encourage active involvement: *"Just this little bit of encouragement, a little bit of change in the way we interact with the (female) students actually encouraged them, and then we saw a lot more participation."* Another possible factor mentioned by Miles [P16], Ivy [P18] and Leo [P14] was their focus on individual support in their classrooms to female students or ensuring their voices are heard. In larger classrooms, where individual attention is infeasible, Ella [P15] and Robin [P12] suggested that they use online tools or Post-it notes to write down their thoughts, which *"allows everyone to contribute (and ...) also pay attention to ensure female students feel seen and heard, encouraging them to share their thoughts."* Nicky [P4] encourages female students but mentions a fine line between encouraging participation and being overbearing. Educators must be mindful of how they encourage participation to avoid making students feel pressured [46].

“From my experience, if you create a friendly environment (in the classroom), female students really like it. - Evan [P17]”

5.2.2 Group Work. Team projects and student groups are indispensable components of CSHE [37]. The collaborative nature of these projects mirrors real-world scenarios where teamwork, communication, and diverse perspectives are crucial for success [5, 37]. Engaging in group work helps students develop skills highly valued in the tech industry, such as problem-solving, project management, and the ability to work effectively in diverse teams [56]. Recognising these benefits, educators frequently (n=12) confirmed the necessity of group work in CSHE. They discussed group work dynamics, emphasising the importance of fair and inclusive group formations to prevent gender-based discrimination and ensure equitable participation. Various strategies were highlighted to achieve these goals.

Bobby [P1] discussed an incident where he had to alter his strategy of students voluntarily forming groups in a team project unit. A group of three male students were apprehensive of including an unallocated female student because that *"would disturb their cool vibe of being in a boys only team."* Thereafter, he adopted a random group formation method to ensure fairness and prevent any potential biases. He stated, *"I changed my method of forming groups for the entire unit... I said, if that's the case, we do split the groups randomly now."* This approach helped in creating diverse teams and encouraged interaction among students from different backgrounds. Alex [P6] mentioned random allocation of groups in his units, as *"Randomly mixing students in groups encourages interaction among diverse students and helps break down social barriers."* Sara [P3] emphasised the value of having female students in groups, even when using random group formation. She noted, *"I do groups randomly. I would go around to a group and say, Oh my God, you got a girl in your group, this is awesome, you will get so much more out of this because you've got different perspectives."*

Similarly, Sara [P3] initially considered placing a female student in every group to provide a female perspective but eventually decided against it. She explained, *"I thought having a girl in every group (...) So, they have a female perspective. But then, conversely, I worry about that. They're the only female in the group. (...) But that's not really fair on them [female students], and they're having to fight, potentially, if they've got some discrimination in the group. So in the end, I decide not to enforce that kind of thing [and form groups randomly], as much as I wanted to, to give them to see, so the guys could see a female perspective."*

Nicky [P4] shared Sara's concerns, stating that assigning one female student per group often leads to issues for female students. On the other hand, Tanya [P8] attempted to discourage all-female groups, as mixed groups with female students in leadership roles tend to perform better (also corroborated by Robin [P12] and Luke [P19]) – also covered in RQ3 (Female engagement).

“Mixed-gender groups tend to work better, with female students often taking leadership roles and excelling in organising tasks and meeting deadlines. - Tanya [P8]”

5.2.3 Pink Tasks. In the context of CSHE, “pink tasks” (the phrase was used by interviewee ()) refer to tasks that require high standards and timeliness but offer little substantive development or visibility, such as documentation or organising meetings [23, 57]. According to our interviewees (n=8), these tasks are often disproportionately assigned to female students, which can perpetuate gender stereotypes and limit their opportunities for technical development.

Ash [P11] shared the most experiences on ‘Pink Tasks’ in her interview and observed that *“All the leadership roles are taken by male students most of the time, and the female students what they do is like documentation, or if there is a small technical task, and they know what to do.”* Such dynamics, where male students preemptively take over tasks, tend to undermine female students’ confidence and autonomy, perpetuating gender stereotypes about capability and roles in CSHE. Ash [P11] recounted another instance where male students took over a task from a female student, stating, *“In one instance, students were tasked with setting up a Raspberry Pi. And then the boys come to set it up for the girl. I asked him, ‘What are you doing?’ And then he said, ‘I’m helping.’ I said, ‘I don’t think she needs help. I’m here to help if she needs it.’ But they still kept on coming, and then the girl was lost [overwhelmed].”* This behaviour affects individual learning experiences and shapes the broader educational environment, reinforcing or challenging existing gender biases [27]. To mitigate this, therefore, Ash [P11] mentioned that *“I try to get some girls to help each other as well rather than getting the boys involved or go and help myself.”*

Instances of gender-based task allocation can extend beyond “pink tasks” to include technical tasks, where female students are often harassed or pressured into sharing their work, or allocated work against their will. Sara [P3] shared, *“Yes, especially in group work, where I’ve seen issues like inter-student plagiarism driven by gender dynamics. Female students, particularly international ones, have been harassed by male teammates to provide their work. To combat this, I made explicit rules about sharing work and emphasised that enabling plagiarism is not helping others.”* Similar incidents were also shared by Dev [P10] and Robin [P12]. The traditional presumed gender roles often emerge in group settings, with female students handling presentations and documentation while male students focus on technical work.

“If there’s a team of students, and there’s one woman, they end up doing more sort of administrative things. - Ella [P15]”

5.3 Student Educator Interaction Dynamics

The approachability, comfort, and authority dynamics between students and educators significantly influence gender inclusivity in CSHE (n=15). We noticed from the interviews that female students benefit from having approachable female educators, however, male students often feel more confident and engaged, especially with male tutors, often at the cost of a lack of respect for female educators’ authority. The systemic issues of respect and authority further exacerbate these challenges, creating an environment that can be unwelcoming for female students and educators.

A significant theme (n=13) was female students’ comfort level when approaching female educators or female educators offering a supportive environment for all students, particularly for female students. All female educators said being authentic and approachable as a teacher is crucial for

building trust and encouraging student engagement, particularly for female students. Ash [P11] mentioned that *“Being a female lecturer makes me more approachable to female students. I try to create a personal connection (...) which helps in building rapport.”* All other female educators, Alex [P6], Max [P7], Evan [P17] and Luke [P19], agreed with this sentiment from Ash [P11]. From the non-female educators, only Alex [P6], Evan [P17] and Luke [P19] emphasised the need to be approachable and maintain an ‘empathetic’ learning environment so that female students engage better and feel more comfortable participating in class discussions.

However, the dynamics shift notably when considering interactions between female educators and male students. Tanya [P8], on the topic of approachability of students, mentioned that *“(female) students might find it easier to approach me with their issues. However, some may try to take advantage of this perceived softness, particularly male students seeking deadline extensions.”* Four other female educators shared the same sentiment that male students assume they are ‘soft target’ and hence easier to push for deadline extensions and get additional marks in assignments, and Ella [P15] echoed this:

“Yes, some students take advantage, expecting leniency from female lecturers compared to males. (...) They think that we are a soft target, so they can be a little bit more pushy.”

All female educators (except Nicky [P4]), Hans [P20] and Alex [P6], reported challenges in maintaining authority and being taken seriously by (male) students. Tai [P5] shared, *“Male students sometimes like to correct me as if to put me down.”* This highlights the persistent gender biases that female educators face, which can undermine their authority and effectiveness in the classroom. These biases are reflected in several facets, e.g., not believing in female educators’ technical know-how, female educators generally receiving lower student feedback scores, getting ‘badgered’ for remarking and extension requests, and not accepting straight answers and counter-questioning their authority much more than they would do for male educators. Hans [P20] and Ash [P11] recounted from their experiences that:

“Female tutors may gain the real respect from students from week 5 (after the semester start), and need a bit more time. (...) Female lecturers often receive relatively low ranking than their male colleagues. - Hans [P20]”

“Female teachers have to work harder to be taken seriously, especially in technical teams. They need to be firm and demonstrate their knowledge to gain respect. - Ash [P11]”

Moreover, Meera [P2], Tai [P5], Alex [P6] and Luke [P19] mentioned that students of other genders generally feel more included and accepted in a learning environment fostered by educators who acknowledge and support their identities in the classrooms. This was also evident in the vocabulary of their interviews, where they consistently used terms such as ‘cis-gender’ or ‘female-appearing students’. It is interesting to note that all other educators when talking about gender aspects, noted only male and female genders on almost all occasions.

“I’m also openly non-binary. Students, particularly some of the queer students are more comfortable to come (to me) and speak about issues, just because I’m open about that as part of who I am. - Alex [P6]”

5.4 Institutional Aspects

Institutional factors, such as hiring practices, training programs, and policies aimed at fostering an inclusive environment, significantly shape gender inclusivity in CSHE (n=10).

One crucial aspect is the gender balance within teaching and support staff. Several educators (n=8) emphasised the importance of employing female tutors and lecturers to create a more balanced educational environment. For instance, Sara [P3] mentioned, *“I’ve always tried to employ lots of*

females in my tutoring... at least half of my tutors, sometimes more than half have been female... that's one of the things I've really tried, to have role model in my course." Similarly, Nicky [P4] stated, *"We try and practice what we preach. So the teaching team that I work on, we make a conscious effort to try and have a 50-50 gender balance [to support female students]."* These efforts ensure that female students see themselves represented in their educators, which can enhance their sense of belonging and encourage their participation in the field.

The presence of women in leadership positions within institutions and the culture fostered can also have a positive impact on the educational culture. Ella [P15] observed, *"Positive changes in hiring practices and culture with more women in senior positions at [our university]. Women in leadership roles can create a more inclusive environment."* This leadership can set a tone for the entire institution, promoting gender inclusivity from the top down. She also noted the importance of workplace culture and the importance of acknowledging the subtle forms of exclusion and bias that female students and staff might face, which can accumulate over time and significantly impact their experience. Microaggressions, though often not overt, can create a hostile or unwelcoming environment, emphasising the need for ongoing efforts to address and mitigate these issues [30].

“It (discrimination) happens often, in subtle ways (at universities). It's not blatant sexism, but small things like being left out of projects (speaking for both staff and students in this context) or not taken seriously. These accumulate and have a significant impact. - Ella [P15]”

Beyond hiring practices, the educators recognised (n=8) that institutional policies and formal training for educators on gender inclusivity, make an impact on inclusive environments. As Bobby [P1] pointed out, *"Lack of formal training for educators on handling gender inclusivity issues (is an key factor)... I strongly believe anybody who starts teaching at this University or any other university should undergo a certain level of training specifically for that."* Training programs, such as university modules on diversity and inclusion, are essential for maintaining an inclusive teaching environment. The same sentiment was echoed by Meera [P2] where she noted that a new staff training module helps - *"There's a University module, and a new faculty module on Diversity and inclusion, which is helpful."* Similarly, Alex [P6] mentioned *"I think that the new inclusivity plan in EDI (by the university) is promising."*

RQ2 Key Findings:

- **External Influences:** Early educational experiences and societal norms significantly shape gender inclusivity in CS. Stereotypes and biases in primary and secondary education often discourage girls from pursuing CS, while cultural and economic factors can either support or hinder female participation in STEM fields.
- **Classroom Dynamics:** Inclusive teaching practices, equitable group work, and careful task assignments are vital for promoting gender inclusivity. Educators must ensure fair and inclusive group formations to prevent gender-based discrimination and support diverse participation.
- **Student-Educator Interactions:** The approachability and supportiveness of educators are critical for encouraging female students' participation and confidence. However, challenges remain in maintaining authority and addressing biases that male students may hold against female educators.
- **Institutional Environment:** Policies and practices related to hiring female faculty, providing support systems for female students, and offering training on gender inclusivity for educators are crucial. A balanced representation of genders among faculty and in leadership positions fosters a more inclusive environment.

6 RQ3 RESULTS

We aimed to find out what different facets of CS students have a major influence on their participation in learning activities, as perceived by their educators. Understanding these facets is crucial for creating an inclusive and effective educational environment. Key themes were *cultural variations* – differences experienced due to culture and country of origin , and *gender* – differences experienced by women and men students. Fig. 3 shows the key RQ3 themes, and Fig. 4 shows how frequently each theme was raised across our 20 educators.

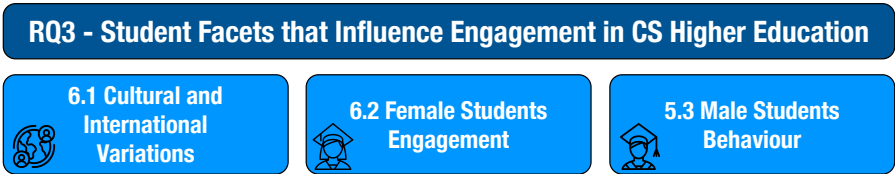


Fig. 3. RQ3 Key Themes

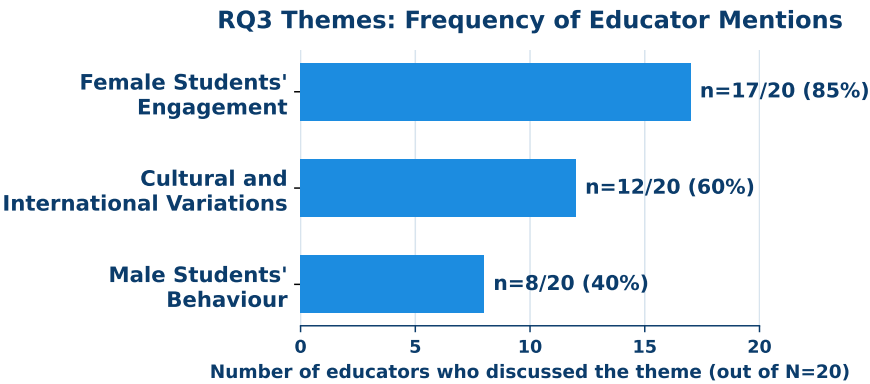


Fig. 4. Number of educators (out of $N = 20$) whose interviews contributed to each RQ3 theme.

6.1 Cultural and International Variations

While some cultural and societal aspects have been addressed in RQ1 and RQ2, this theme specifically explores how students' cultural backgrounds, particularly international students, influence their engagement in CSHE classrooms, particularly emphasising gender dynamics. The interviews revealed several facets of this influence ($n=12$), particularly concerning international students who often come from educational environments vastly different from those in *Anon_Region*.

Firstly, there is a notable trend of women students in master's courses coming from overseas. Several educators, e.g., Bobby [P1], Max [P7], Robin [P12], Andy [P13], and Ivy [P18], highlighted that a majority of women students at this level are international, which underscores the cultural diversity within the classroom. This diversity brings both opportunities and challenges in terms of participation and engagement.

One of the challenges is the adaptation period required for international students to acclimatise to their new educational and cultural environment. Tanya [P8] mentioned that this adaptation

period is significant, sharing an example of an international Muslim women student who was apprehensive about working with male students due to cultural norms. Miles [P16] noted similar trends for students from Middle Eastern countries, where he mentioned that male students do not interact with women tutors and peers, and similarly, women students are apprehensive about interacting with male tutors and peers. Luke [P19] mentioned that he often encounters 'Asian' men students who often go to boys-only schools and then find it difficult to adapt to their new learning environment. This indicates that cultural sensitivity and support are crucial in helping these students feel comfortable and integrated [8].

Tai [P5] echoed this sentiment, observing that male international students, especially from China, potentially due to their cultural context, do not take her seriously (similar to student-teacher interaction in RQ2), and a substantial effort is required to establish her authority and support the students in their learning process.

“Someone from the Middle East might have certain preferences. For example, Muslim men are (sometimes) not comfortable going to a female tutor, and the other way around, female Middle Eastern students have similar preferences. - Miles [P16]”

Furthermore, international students often exhibit different classroom behaviours based on their cultural upbringing. Sam [P9] and Ivy [P18] noted that many international students, particularly from countries with hierarchical educational systems, tend to be shy and avoid asking questions. Ivy [P18] explained that these students are taught to be obedient and not question authority figures, which can hinder their active participation in class. Tai [P5] echoed this sentiment, observing that male Chinese students rarely question anything. This lack of engagement can be a barrier to their learning and integration into the classroom environment.

However, Ash [P11] provided a contrasting perspective, stating that international students often perform better academically once they overcome initial cultural and educational barriers. This suggests that while these students may initially struggle with participation, they are capable of excelling academically with the right support and encouragement.

The issue of plagiarism also surfaced in the context of cultural aspects. Sara [P3] and Ash [P11] reported instances where male international students sought help from women students and assigned them “pink tasks” such as documentation. This not only placed an unfair burden on women students but also perpetuated gender stereotypes and inhibited the development of technical skills among women students.

Overall, these cultural aspects highlight the importance of understanding and addressing the diverse needs of international students, who are more than 50% cohort in postgraduate CSHE classrooms, to foster a more inclusive and supportive educational environment.

These accounts are themselves an intersectional analysis of gender, even where educators did not use that language. The pattern educators describe is not ‘international students behave differently’ or ‘women behave differently’ treated separately, but a joint effect of gender *and* cultural/religious background: Miles’ and Tanya’s observations concern specifically Muslim women’s and men’s preferences for tutors of a particular gender, and Luke’s and Tai’s observations concern specifically male students from particular educational cultures being less accustomed to women in positions of authority. In both cases, the reported behaviour is contingent on the combination of gender and cultural background, not attributable to either alone – precisely the kind of co-constitution that intersectionality theory foregrounds [14], and that calls for moving beyond gender as a single-axis category when interpreting participation and engagement in culturally diverse CSHE cohorts [22, 45]. At the same time, we note a methodological tension consistent with our Limitations (Section 8): because we did not collect or report individual-level race/ethnicity or nationality to protect participant anonymity, we can characterise these intersections only at the level of the

patterns educators describe in aggregate, not verify them against participants' own layered identities. We return to this tension, and what it implies for future intersectional research designs, in Section 7.

6.2 Female Students' Engagement

Several educators contributed to this theme (n=17). Educators mentioned that women students are generally more engaged in classroom activities than male students, bringing higher dedication and thoroughness. Many top-performing students in classes are females, indicating their strong commitment and motivation. We discuss the educators' viewpoints on women students, wherein educators mention that women students are dedicated, engaged, willing to learn from feedback, responsible, excellent leaders, produce high-quality outputs, and have clarity of thought, but also shy, less likely to volunteer in discussions, and less likely to directly ask questions in the classroom.

Max [P8] commented, *"Many top students in my classes are females because they are dedicated and thorough."* Evan [P17] remarked, *"Among the top students, there's always a cohort of females who are highly motivated and curious, wanting to know more and really understanding why they are there, which helps them to stand out."* As observed by Tai [P5], Ash [P11] and Robin [P12], women students generally understand, effectively learn from feedback, and contribute better to the learning environment. Tai compared women students to male students, and commented that *"Female students generally understand and learn from the feedback. Male students often get defensive and sometimes escalate the issue to male staff, hoping for leniency."*

Female students often take on leadership roles in group projects, excelling in organising tasks and meeting deadlines. Ella [P15], Tanya [P8], Hans [P20], Ash [P11] mentioned, in mixed-gender groups, female students usually contribute significantly to the project's overall success. For example, in a current capstone project, a female student led a group of six, including four males, and is performing exceptionally well. Tanya [P8] noted *"Female students often take on leadership roles (in mixed gender groups) and excel in organising tasks and meeting deadlines."* Ella [P15] mentioned that although, in her experience, it's not common for women students to take on a leader's role in teamwork, but *"when it happens, those teams are often very cohesive and well-organized."*

Female students are less likely to badger for extra marks compared to their male counterparts, who often wait for assignments and may seek grade adjustments. One educator pointed out, *"Female students are likely to not badger for extra marks, as compared to male students."* This indicates a more consistent and responsible approach to their academic responsibilities.

The overall perception of women students' intelligence and capability is positive, with educators highlighting their understanding and adaptability. Evan [P17] emphasised the importance of a welcoming environment, *"Females are smarter compared to the male cohort is always there. They understand things. The only thing I would say is to make things welcoming for the female."* Female students often show proactivity and clear career goals, particularly in IT professions. This determination is reflected in their academic performance and leadership qualities.

“Female students enter an IT profession most times knowing exactly what they want, whereas the males can come there just because a friend suggested it, or they think it might be easy, or because they enjoy video games. - Luke [P19]”

However, women students are sometimes 'shy' in the classrooms and hesitant to volunteer for activities, particularly in technical or mathematical contexts (according to Bobby [P1] and Sam [P9]). Bobby [P1] mentioned that they may need additional encouragement and individual feedback, which can help increase their participation and confidence. Female students may prefer less public forms of engagement, such as asking questions after class or via email. This indicates a preference for a supportive and encouraging classroom environment. Sara [P3] noted, *"Female students may*

be more likely to ask questions after class or via email rather than in front of the entire class. This might indicate a preference for less public forms of engagement.”

In conclusion, female students’ engagement in CSHE is marked by their dedication, leadership abilities, and responsible approach to academic tasks. Their contributions enrich the learning environment and highlight the importance of fostering a supportive atmosphere. However, they might have some barriers such as not asking questions or engaging in public discussion, that can be overcome with individual attention and encouragement.

6.3 Male Students’ Behaviour

Most educators did not mention anything specific related to male students, except some educators (n=8) noted that male students tend to participate more actively in classroom discussions, possibly due to their greater numbers. Certain specific facets mentioned in the interviews by some educators (e.g., Robin [P12], Miles [P16] and Tanya [P8]) about male students include their tendency to focus more on assessment than learning, e.g., “badgering for extensions” or more marks. Some educators shared a few anecdotes about male students’ behaviour in CSHE classrooms, which presents unique challenges and dynamics that impact overall classroom engagement and inclusivity, which we discuss below.

As also covered in RQ2 Group Work - male students’ resistance to working in groups with women students often necessitates intervention from educators to ensure fair and equitable group formation. Bobby [P1] noted, *“Male students sometimes resist working in groups with female students, which requires intervention from educators to ensure fair and equitable group formation.”* Robin [P12] also added *“Male students typically prefer working with their peer group, often consisting of other males, especially those they know from past experiences.”* Group work dynamics also reveal issues related to gender-based harassment and task allocation.

“In group work, where I’ve seen issues like inter-student plagiarism driven by gender dynamics. Female students, particularly international ones, have been harassed by male teammates to provide their work. To combat this, I made explicit rules about sharing work and emphasised that enabling plagiarism is not helping others. - Sara [P3]”

Addressing biases and stereotypes is another significant challenge with male students. Educators have observed that male students sometimes create stereotypical characters and exhibit casual sexism and racism, creating an uncomfortable environment, particularly in gaming-related units. Both Tai [P5] and Alex [P6] noted that a key challenge is teaching male students to build non-stereotypical characters for their game design assignments. Alex [P6] noted *“Casual sexism among (male) students can create an uncomfortable environment. Addressing these issues immediately and fostering open discussions are crucial for maintaining inclusivity.”* Alex [P6] further noted, *And the way that I found to deal with that is to not... attack them directly... but I always use it as a launching point for a discussion.”* Tai [P6] shared that sometimes students would just avoid any female characters in the game and say “we avoided sexism”, or build female characters with “pronounced bodyparts”, “damsel in distress”, “named wife” and just “get them killed” unless “a prince comes and saves them”. Tai [P6] observed that male students often misunderstand or resist any efforts for non-stereotypical characters. This resistance can be particularly pronounced when students are penalised for creating stereotypical characters, leading to frustration and defensiveness. Tai [P5] and Alex [P6] often aimed to encourage male students to develop a more inclusive and respectful perspective, but these efforts can be met with pushback. Tai [P6] shared an anecdote on how a male student complained about her to a senior colleague as a pushback saying *“It’s a gaming class - not a feminism class.”*

Despite these challenges, male students can significantly support their female teammates by encouraging them to take on leadership roles and technical tasks. Positive dynamics have been

observed in some teams where male students support and recognize female leaders. Ella [P15] shared, “Male students can play a significant role in supporting their female teammates by encouraging them to take on leadership roles and technical tasks. Some teams exhibit positive dynamics where male students support and recognise female leaders.”

Overall, male students’ behaviour in CSHE classrooms highlights the need for continuous efforts to address gender biases, promote inclusivity, and create a supportive environment for all students.

RQ3 Key Findings:

- **Cultural and International Variations:** International students bring diverse cultural backgrounds to the CSHE classrooms, which impacts their engagement. Students’ apprehension with tutors of different genders, the adaptation period and cultural sensitivity are critical for their full participation. Challenges include initial shyness, reluctance to ask questions, and differences in classroom behaviours, which necessitate tailored support and inclusive practices from educators.
- **Female Students’ Engagement:** Female students generally exhibit higher dedication, thoroughness, and leadership qualities in classroom activities. They often excel in organising tasks, meeting deadlines, and performing well in mixed-gender groups. However, they may be less likely to volunteer in discussions and prefer private forms of engagement, such as after-class queries. Providing a supportive and encouraging environment is essential to enhance their participation and confidence.
- **Male Students’ behaviour:** Male students tend to be more vocal and active in classroom discussions however, may exhibit resistance to working with female peers and creating non-stereotypical characters in assignments. Issues such as casual sexism, grade disputes, and task allocation biases can create an uncomfortable environment. Educators need to address these biases through open discussions and promote positive dynamics where male students support their female teammates.

7 RECOMMENDATIONS

The findings from our RQs provide critical insights into the experiences and challenges faced by women and non-binary students in CSHE, based on educators’ perspectives. Based on these insights, we present a set of comprehensive recommendations, based on the recommendations from our interviewees, aimed at various stakeholders, including educators, institutions, and researchers. These recommendations are designed to foster a more inclusive and supportive academic environment, address systemic barriers, define future research directions, and promote gender diversity in CS education. Fig. 5 shows our key recommendations for CSHE educators and institutes, and researchers. Table 4 presents a mapping of the recommendations for CS educators and institutes from the responses we received. The key recommendations are made directly from the responses as well as from the synthesis of the responses. Hence, the recommendations below are a combination of the explicitly mentioned recommendations by the educators and us. By implementing these strategies, we can work towards creating a more equitable and dynamic educational landscape that benefits all students. The following subsections outline targeted actions for educators and institutions to directly impact classroom practices and institutional policies and research directions to further our understanding and improve gender inclusivity in CS education.

Before detailing individual recommendations, we make explicit how our findings speak back to the three theoretical lenses introduced in Section 2, since the recommendations below are, in effect, an attempt to intervene on the mechanisms those theories identify.

Stereotypes, field culture, and ambient belonging. Our findings corroborate this lens directly: educators repeatedly described CS and its more technical sub-fields (e.g., cybersecurity, hardware) as culturally coded as a ‘boys’ club’ (Section 4), and linked this coding to ambient cues in the classroom, in media, and in early schooling that signal who computing is ‘for’ [9, 10]. The recommendations on representation, gender-inclusive language, and diverse case studies (below) are best understood as attempts to change these ambient cues rather than as generic ‘good teaching’ advice.

Expectancy-value and social-cognitive accounts of choice. Our study did not measure self-efficacy or expectancy beliefs directly, but educators’ accounts are consistent with these constructs operating as an intervening mechanism between stereotypes and participation. Read together with the stereotype and ambient-belonging literature in Section 2, a plausible mechanism is that when CSHE environments signal (through language, examples, or peer climate) that the field is not ‘for’ certain students, expectancy-value and social-cognitive accounts predict this should depress self-efficacy and perceived value before it affects persistence – which is consistent with the pattern educators in our study describe when discussing early discouragement (Section 4.2) and confidence-building strategies in the classroom (Section 5). For example, educators described specific instances in which low-key, individualised encouragement measurably increased female students’ classroom participation (Section 5), which is what an expectancy-value account would predict if the barrier were partly about perceived likelihood of success rather than interest or ability [20, 36, 38]. This reframes several of our recommendations (e.g., personalised feedback, low-stakes participation channels) as expectancy-building interventions, not just inclusivity ‘nice-to-haves’ – and suggests that their success should ultimately be evaluated against confidence and self-efficacy outcomes, not only enrolment or retention numbers.

Identity, belonging, and intersectionality. Our findings show that gender inclusivity in CSHE cannot be addressed as a single-axis problem. Educators’ own accounts of authority, comfort, and belonging were consistently shaped by the intersection of gender with sub-field, religion/cultural visibility, sexuality, and nationality (Sections 4 and 6), consistent with calls to treat intersectionality as a structural analytic in computing research rather than an additional demographic variable [22, 45]. A direct implication is that ‘gender-neutral’ interventions designed without attention to these intersections risk being effective for some students and ineffective, or even alienating, for others. We translate this into concrete recommendations below, and into a dedicated research recommendation on intersectional methods.

7.1 Recommendations for CS Educators and Institutes

Early Intervention and Outreach: Based on our findings, engagement and motivation at school levels are key factors for improving gender inclusivity in CSHE. CS educators and institutes should regularly organise engagement activities, such as school visits and workshops, summer camps, and after-school programs that introduce girls to CS and STEM through engaging, hands-on activities like coding basics and robotics [34]. As mentioned by several educators in our study, open days are an excellent opportunity to demonstrate representation and engage with parents and younger siblings of prospective students. Further, establishing mentorship programs and providing opportunities for students to connect with female and non-binary role models in these fields can inspire and guide them and help with both recruitment and retention [17]. Additionally, advocating for the integration of technology and coding into the standard curriculum and ensuring schools have access to necessary resources, e.g., supported by local universities, will promote early exposure to technology and address the ‘pipeline issue’ of universities.

Inclusive Teaching Practices and Classroom Experience: The other key factor discerned in our study is the importance of inclusive classroom practices and an inclusive environment. On this account, the first and most straightforward step is using gender-inclusive language. Educators

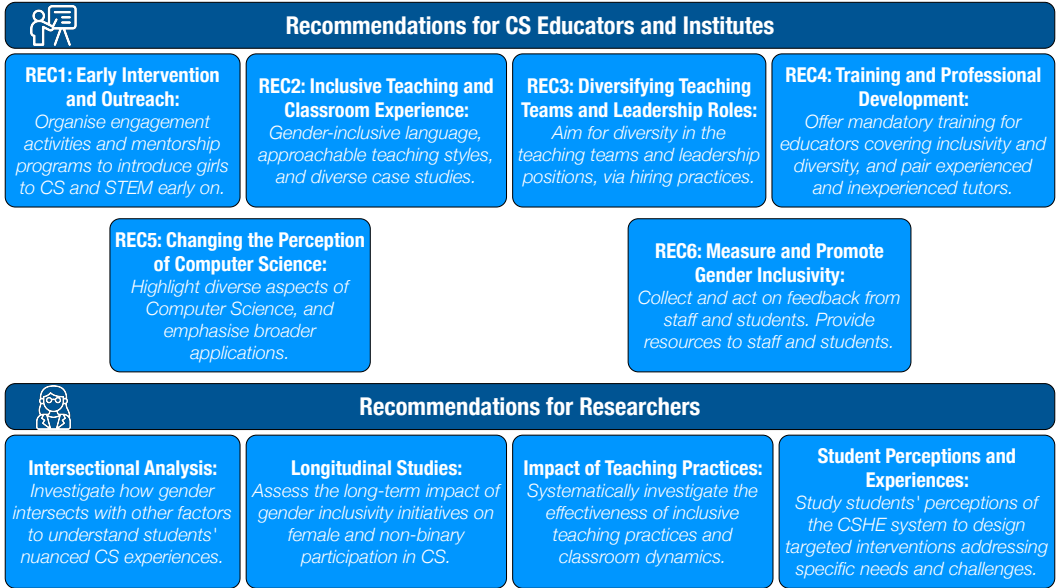


Fig. 5. Recommendations

Table 4. Mapping: Educators and Recommendations

Educator	REC1	REC2	REC3	REC4	REC5	REC6
Bobby [P1]	✓	✓		✓		
Meera [P2]		✓	✓	✓		✓
Sara [P3]		✓	✓			
Nicky [P4]		✓	✓			
Tai [P5]		✓		✓	✓	✓
Alex [P6]		✓	✓			✓
Max [P7]	✓		✓			
Tanya [P8]	✓					
Sam [P9]						
Dev [P10]	✓		✓			
Ash [P11]	✓		✓			
Robin [P12]	✓		✓		✓	
Andy [P13]	✓				✓	
Leo [P14]	✓				✓	
Ella [P15]			✓		✓	
Miles [P16]		✓		✓		✓
Evan [P17]		✓	✓		✓	✓
Ivy [P18]		✓	✓			✓
Luke [P19]	✓	✓				
Hans [P20]	✓			✓	✓	

should consistently use gender-neutral language in all classroom communications, materials, and assessments to foster an inclusive environment [58]. At an institute level, it is important to conduct inclusivity training modules for teaching staff members to understand and implement inclusive practices, as indicated by several educators in our study. The course content should include diverse case studies, examples, and role models from various genders, races, and backgrounds to reflect a

wide range of perspectives and experiences. This approach can be integrated into teaching and assessment materials. By showcasing diverse contributors to the field, educators can help students from underrepresented groups see themselves in these roles, thereby promoting a more inclusive learning environment [50]. Creating a supportive and approachable atmosphere where students feel comfortable seeking help and discussing issues is essential. Educators should build strong relationships with their students, offering personalised feedback and encouragement, and institutes can support such practices by offering appropriate workload support to educators for personalised teaching and feedback.

Diversifying Teaching Teams and Leadership Roles: Striving for a balanced representation of genders among faculty and in leadership positions is essential for creating an inclusive environment [3]. Our educators consistently highlighted the need for diverse representation among educators regarding gender. Institutions should actively recruit and support female and non-binary faculty members, providing mentorship and professional development opportunities to help them succeed. Developing robust support systems for female faculty members is critical for their retention and success. Institutions should offer networking opportunities, professional development, and leadership training tailored to the unique challenges faced by female and non-binary faculty. These opportunities should be beyond token representation and not targeted at “*getting diversity in different meetings and panels*” - Ella [P15]. Mentorship programs that pair junior faculty with experienced mentors can provide guidance and support in navigating their academic careers [3, 35].

Training and Professional Development: Training and professional development are critical for promoting inclusivity and diversity in CSHE. Educators highlighted the importance of offering mandatory training workshops for educators to address gender inclusivity, implicit bias, and effective teaching practices. Such training should include practical case studies and promote active engagement, enabling educators to apply inclusive strategies in the classroom [21]. Some educators suggested pairing experienced educators with inexperienced ones can further foster a culture of mentorship and skill development, ensuring that all educators are well-prepared to handle diverse student needs. Professional development initiatives should also teach educators to create welcoming environments, use gender-neutral language, and handle challenging situations sensitively. By investing in training and professional development, CS programs can cultivate teaching teams that support all students equitably and foster an inclusive culture that empowers underrepresented groups.

Changing the Perception of Computer Science: Changing the perception of CS is crucial for broadening participation and fostering inclusivity. It is related to REC1 as well, as one way of addressing this is via engagement at a school level. Our educators emphasised the importance of highlighting the diverse aspects of CS beyond just programming, such as its applications in social sciences, business, and creative fields [44]. This approach can help challenge the prevailing stereotypes that depict CS as a field only for “geeks” or those with a highly technical background. Evan [P17] mentioned the importance of engagement with media for CS academics and promoting the varied and interdisciplinary nature of CS. It can attract students who may otherwise feel the field is not suited to their interests or skills. Emphasising broader applications and skills like communication, teamwork, and problem-solving can make CS more accessible, particularly to women and other underrepresented groups. By reframing CS as a versatile field with real-world impact, educators and institutions can cultivate a more diverse and engaged cohort of students.

Measure and Promote Gender Inclusivity: Some educators emphasised the importance of collecting regular feedback from both staff and students to identify gaps and biases in the teaching process. This ongoing feedback can help educators refine their approaches and make necessary adjustments to foster inclusivity. Another key aspect of collaborative projects and group work is essential in CSHE. Educators should focus on forming diverse groups for collaborative projects and

monitor group dynamics to prevent gender-based discrimination and ensure fair task distribution. Educators should assign roles within groups to avoid the common pitfall of women students being relegated to administrative tasks (“pink tasks”) while male students take on more technical roles [23]. Regular check-ins with groups can help ensure that all students contribute equally and benefit from the group work experience. At an institute level, the educators need support and resources from their institutes for initiatives promoting gender inclusivity in CSHE. This can include financial support for outreach programs, mentorship schemes, and professional development opportunities focused on inclusive teaching practices. Creating grants and scholarships aimed at supporting female and non-binary students in CS can help reduce financial barriers, provide additional motivation and encourage greater participation [35].

7.2 Future Work and Recommendations for Researchers

Intersectional Analysis: As discovered in our findings, gender is a major, yet only one aspect for improving diversity in CSHE. As researchers, we need to further investigate how gender intersects with other factors such as race, socioeconomic status, and cultural background to understand students’ nuanced CS experiences. This involves conducting studies that consider multiple axes of identity and their combined impact on students’ educational journeys. By examining these intersections, researchers can uncover specific barriers and challenges faced by different groups, which may not be apparent when looking at gender in isolation. For example, the experiences of a woman of colour in a CS program might differ significantly from those of her white peers due to additional layers of discrimination and bias [6]. Our own study illustrates why this is difficult in practice: as discussed in Section 8, we deliberately did not collect or report individual-level race/ethnicity or nationality because our sample size and recruitment method made re-identification a realistic risk, which meant we could only interpret intersections at the level of aggregate patterns rather than individual layered identities. Future work should treat this as a design problem rather than an afterthought: larger, multi-site samples that permit k -anonymity when reporting demographic combinations, participatory or community-based methods that give participants control over how much of their identity is disclosed and to whom, and reporting conventions that separate analytic categories from directly identifying detail, would all allow researchers to move closer to the structural, rather than additive, treatment of intersectionality called for by Rankin and Thomas [45] and Erete et al. [22]. We see this as a direct, practical next step from our own Limitations rather than a generic call for ‘more intersectional research’.

Longitudinal Studies: This study attempted to understand the educators’ perceptions of gender inclusivity in CSHE settings. Longitudinal studies are required to assess the long-term impact of gender inclusivity initiatives and early interventions on female and non-binary participation in CS. By tracking cohorts of students over multiple years, researchers can evaluate how specific programs and policies influence enrollment, retention, and career progression. This approach provides valuable insights into the sustainability and effectiveness of the efforts to develop gender inclusive CSHE environment.

Impact of Teaching Practices: Teaching practices and classroom environment were among the key factors that influence gender inclusivity, according to our findings. There is a need to systematically investigate the effectiveness of various inclusive teaching practices and classroom dynamics to identify best practices and develop evidence-based guidelines for educators. This includes analysing the impact of gender-neutral language, diverse representation in course materials, and strategies for encouraging active participation and equitable group work. In such studies, we recommend researchers to look for correlations between teaching practices and student outcomes, such as academic performance, engagement, and retention. Because our interview data can only describe educators’ perceptions of *what* works, not measure *why*, we specifically recommend that follow-up

studies incorporate validated expectancy-value and self-efficacy instruments [20, 36, 38] alongside outcome data, so that researchers can test whether the inclusive practices educators describe (e.g., individualised encouragement, low-stakes participation channels) operate through the mechanisms these theories predict, rather than assuming the link.

Student Perceptions and Experiences: This study focused on educators’ perceptions and experiences of gender inclusivity. There is still a gap in systematically studying students’ perceptions of the CSHE environment with diverse experiences of students from different backgrounds. As researchers, we should examine disparities in experiences and outcomes among different student groups. Identify the factors that contribute to positive or negative experiences (similar to our RQ2) and use this information to design interventions that address specific needs and challenges (and map them against findings of RQ3 in our study from educators’ perspective).

8 LIMITATIONS AND THREATS TO VALIDITY

There are a few potential threats that could limit our findings. Representativeness of the sample is always a concern with convenience sampling and relatively small-scale qualitative research. While convenience sampling can inherently limit generalisability, the interviewees in our study had a wide range of experiences in CSHE contexts, from 4 to 40 years, ensuring a breadth of perspectives [15]. Moreover, the richness and diversity of the experiences reported in the interviews suggest that our sample was sufficiently varied to provide meaningful insights. The qualitative analysis was conducted iteratively, and data saturation was reached after approximately 16 interviews, with four additional interviews yielding no new themes, further validating the comprehensiveness of our findings. We note that our findings can only be interpreted as situated within the study context and not be read as evidence that gender disparity in CSHE manifests uniformly across global regions.

A related threat is *self-selection and network-related sampling bias*. Participants were recruited via email outreach through professional networks, and participation was voluntary. This may bias the sample towards educators who are more willing to discuss gender inclusivity or who have stronger interest in the topic. We acknowledge that this may influence which perspectives are represented, and we therefore frame our claims as interpretive findings grounded in this dataset rather than as estimates of prevalence. Another threat relates to *how participants expressed themselves during interviews*. As gender inclusivity is a subjective and potentially sensitive topic, participants may have been influenced by social desirability bias, used high-level or normative language (i.e., what is “expected”), or avoided discussing specific incidents due to confidentiality concerns. Further, differences in communication styles and prior experiences (including experiences outside the Western context) may have shaped how participants articulated their perspectives. To mitigate these issues, we used semi-structured prompts grounded in lived experiences (e.g., asking for recalled experiences and examples), encouraged participants to speak in general terms without identifying individuals, and used follow-up probes to clarify meaning and reduce ambiguity.

A further threat concerns *researcher interpretation and coding bias*. *Qualitative thematic analysis necessarily involves interpretive judgement, and researchers’ prior assumptions can influence code creation, theme boundaries, and the emphasis placed on particular excerpts. To mitigate this, the first author conducted primary coding, the second author independently reviewed and validated the initial code set, and themes were reviewed and refined through discussion with the second and third authors. We maintained an audit trail of code merges/renames and theme revisions to support traceability from excerpts to codes and themes. We focused on consistency by independently reviewing initial codes, removing and merging duplicates, and consensus-based refinement of code definitions, which were then applied across the full dataset.*

Finally, because we could not safely report certain potentially identifying participant attributes (e.g., academic role, race/ethnicity, or nationality) at an individual level without increasing re-identification risk, we did not conduct a layered intersectional analysis beyond gender. This limits our ability to examine how gender inclusivity may interact with other identity dimensions. Another possible threat is interviewees sharing responses that are unrelated to our RQs. Since the data collection format was interviews, the interviewer carefully facilitated the conversations in the context. We also prepared some “transition questions” that would be useful for facilitating the conversation and breaking the ice, if needed. We also used the semi-structured interview guide to keep the discussion aligned with our RQs, and asked clarification questions when responses were broad or ambiguous.

9 CONCLUSION

This paper draws on the experiences and perceptions of educators on gender inclusivity in Computer Science Higher Education (CSHE). The study explores the persistent gender imbalance in CS classrooms, attributing it to societal stereotypes, early educational influences, and institutional practices. Through qualitative interviews with educators, our study identified key factors that impact gender inclusivity, including the importance of early interventions, inclusive teaching practices, and supportive institutional policies. The paper also highlights the differing engagement patterns and behaviours of male and women students, noting that while women students often exhibit strong organizational skills and leadership qualities, they may be less vocal in classroom settings compared to their male counterparts. Based on these findings, we offer actionable recommendations for educators, institutions, and researchers to foster a more inclusive academic environment in CSHE. The recommendations emphasize the need for early outreach programs, gender-neutral language, diverse representation in course content, and the importance of addressing biases and promoting positive group dynamics in the classroom. This research contributes to the ongoing discourse on gender inclusivity in CSHE by comprehensively analysing the challenges and proposing strategies to create a more equitable and supportive educational landscape.

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REFERENCES

- [1] [n. d.]. University enrolment and completion in STEM and other fields. <https://www.industry.gov.au/publications/stem-equity-monitor> Australian Department of Industry, Science and Resources.
- [2] Sylvia Beyer. 2014. Why are women underrepresented in Computer Science? Gender differences in stereotypes, self-efficacy, values, and interests and predictors of future CS course-taking and grades. *Computer Science Education* 24, 2-3 (2014), 153–192.
- [3] Diana Bilimoria, Simy Joy, and Xiangfen Liang. 2008. Breaking barriers and creating inclusiveness: Lessons of organizational transformation to advance women faculty in academic science and engineering. *Human Resource Management: Published in Cooperation with the School of Business Administration, the University of Michigan and in Alliance with the Society of Human Resources Management* 47, 3 (2008), 423–441.
- [4] Virginia Braun and Victoria Clarke. 2006. Using Thematic Analysis in Psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101. <https://doi.org/10.1191/1478088706qp063oa>
- [5] Andreas Breiter, Görschwin Fey, and Rolf Drechsler. 2005. Project-based learning in student teams in computer science education. *Facta universitatis-series: Electronics and Energetics* 18, 2 (2005), 165–180.
- [6] Heidi B Carlone and Angela Johnson. 2007. Understanding the science experiences of successful women of color: Science identity as an analytic lens. *Journal of Research in Science Teaching: The Official Journal of the National Association for Research in Science Teaching* 44, 8 (2007), 1187–1218.
- [7] Randolph C. H. Chan. 2022. A Social Cognitive Perspective on Gender Disparities in Self-Efficacy, Interest, and Aspirations in Science, Technology, Engineering, and Mathematics (STEM): The Influence of Cultural and Gender Norms. *International Journal of STEM Education* 9, Article 37 (2022). <https://doi.org/10.1186/s40594-022-00352-0>

- [8] Alicia Fedelina Chávez and Susan Diana Longerbeam. 2023. *Teaching across cultural strengths: A guide to balancing integrated and individuated cultural frameworks in college teaching*. Taylor & Francis.
- [9] Sapna Cheryan, Allison Master, and Andrew N. Meltzoff. 2015. Cultural Stereotypes as Gatekeepers: Increasing Girls' Interest in Computer Science and Engineering by Diversifying Stereotypes. *Frontiers in Psychology* 6 (2015), 49. <https://doi.org/10.3389/fpsyg.2015.00049>
- [10] Sapna Cheryan, Victoria C. Plaut, Paul G. Davies, and Claude M. Steele. 2009. Ambient Belonging: How Stereotypical Cues Impact Gender Participation in Computer Science. *Journal of Personality and Social Psychology* 97, 6 (2009), 1045–1060. <https://doi.org/10.1037/a0016239>
- [11] Sapna Cheryan, Sianna A. Ziegler, Amanda K. Montoya, and Lily Jiang. 2017. Why Are Some STEM Fields More Gender Balanced Than Others? *Psychological Bulletin* 143, 1 (2017), 1–35. <https://doi.org/10.1037/bul0000052>
- [12] Victoria Clarke and Virginia Braun. 2017. Thematic analysis. *The journal of positive psychology* 12, 3 (2017), 297–298.
- [13] J McGrath Cohoon and William Aspray. 2006. A critical review of the research on women's participation in postsecondary computing education. (2006).
- [14] Kimberlé Crenshaw. 1989. Demarginalizing the Intersection of Race and Sex: A Black Feminist Critique of Antidiscrimination Doctrine, Feminist Theory, and Antiracist Politics. *University of Chicago Legal Forum* 1989, 1 (1989), 139–167.
- [15] Martyn Denscombe. 2017. *The good research guide: For small-scale social research projects*. McGraw-Hill Education (UK).
- [16] Digest of Education Statistics [n. d.]. Degrees in computer and information sciences conferred by postsecondary institutions, by level of degree and sex of student: 1964–65 through 2018–19. https://nces.ed.gov/programs/digest/d20/tables/dt20_325.35.asp National Centre of Education Statistics.
- [17] Benjamin J Drury, John Oliver Siy, and Sapna Cheryan. 2011. When do female role models benefit women? The importance of differentiating recruitment from retention in STEM. *Psychological Inquiry* 22, 4 (2011), 265–269.
- [18] Jie Du and Hayden Wimmer. 2019. Hour of Code: A study of gender differences in computing. *Information Systems Education Journal* 17, 4 (2019), 91.
- [19] Alan Durnell, Gerda Siann, and Peter Glissov. 1990. Gender differences and computing in course choice at entry into higher education. *British Educational Research Journal* 16, 2 (1990), 149–162.
- [20] Jacquelynne S. Eccles and Allan Wigfield. 2000. Expectancy–Value Theory of Achievement Motivation. *Contemporary Educational Psychology* 25, 1 (2000), 68–81. <https://doi.org/10.1006/ceps.1999.1015>
- [21] Chima Abimbola Eden, Onyebuchi Nneamaka Chisom, and Idowu Sulaimon Adeniyi. 2024. Cultural competence in education: strategies for fostering inclusivity and diversity awareness. *International Journal of Applied Research in Social Sciences* 6, 3 (2024), 383–392.
- [22] Sheena Erete, Yolanda A. Rankin, and Jakita O. Thomas. 2023. A Method to the Madness: Applying an Intersectional Analysis of Structural Oppression and Power in HCI and Design. *ACM Transactions on Computer-Human Interaction* 30, 2, Article 24 (2023), 45 pages. <https://doi.org/10.1145/3507695>
- [23] Rita Garcia, Chieh-Ju Liao, Ariane Pearce, and Christoph Treude. 2022. Gender influence on communication initiated within student teams. In *Proceedings of the 53rd ACM Technical Symposium on Computer Science Education-Volume 1*. 432–438.
- [24] Alicia García-Holgado, Andrea Vázquez-Ingelmo, and Francisco José García-Peñalvo. 2024. Gender Mainstreaming and Diversity in Higher Education: An Experience in Software Engineering Education. *IEEE Transactions on Education* (2024).
- [25] Google CS Ed Research group et al. 2014. Women who choose computer science—what really matters: The critical role of encouragement and exposure. *Tekninen raportti. Saatavana elektronisesti* (2014).
- [26] John Grundy, Tanjila Kanij, Rashina Hoda, Hourieh Khalajzadeh, Anuradha Madugalla, and Jennifer McIntosh. [n. d.]. ED&I and SE: challenges, progress and lessons. ([n. d.]).
- [27] Kai Hakkarainen and Tuire Palonen. 2003. Patterns of female and male students' participation in peer interaction in computer-supported learning. *Computers & Education* 40, 4 (2003), 327–342.
- [28] Google Inc. & Gallup Inc. 2016. Diversity gaps in computer science: Exploring the underrepresentation of girls, blacks and hispanics.
- [29] Letizia Jaccheri, Cristina Pereira, and Svetlana Fast. 2020. Gender issues in computer science: lessons learnt and reflections for the future. In *2020 22nd International Symposium on Symbolic and Numeric Algorithms for Scientific Computing (SYNASC)*. IEEE, 9–16.
- [30] Stephanie Suzanne Judson. 2014. *Sexist discrimination and gender microaggressions: an exploration of current conceptualizations of women's experiences of sexism*. The University of Akron.
- [31] Tanjila Kanij, John Grundy, and Jennifer McIntosh. 2024. Enhancing understanding and addressing gender bias in IT/SE job advertisements. *Journal of Systems and Software* (2024), 112169.

- [32] Päivi Kinnunen, Matthew Butler, Michael Morgan, Aletta Nylen, Anne-Kathrin Peters, Jane Sinclair, Sara Kalvala, and Erkki Pesonen. 2018. Understanding initial undergraduate expectations and identity in computing studies. *European Journal of Engineering Education* 43, 2 (2018), 201–218.
- [33] Yekaterina Kovaleva, Ari Happonen, Jussi Kasurinen, and Eneli Kindsiko. 2023. State-of-the-art review on current approaches to female inclusiveness in software engineering and computer science in higher education. *IEEE Access* (2023).
- [34] Antti-Jussi Lakanen. 2016. *On the impact of computer science outreach events on K-12 students*. Ph.D. Dissertation. University of Jyväskylä.
- [35] Sandra Laursen and Ann E Austin. 2020. *Building gender equity in the academy: Institutional strategies for change*. Johns Hopkins University Press.
- [36] Robert W. Lent, Steven D. Brown, and Gail Hackett. 1994. Toward a Unifying Social Cognitive Theory of Career and Academic Interest, Choice, and Performance. *Journal of Vocational Behavior* 45, 1 (1994), 79–122. <https://doi.org/10.1006/jvbe.1994.1027>
- [37] Robert Lingard and Shan Barkataki. 2011. Teaching teamwork in engineering and computer science. In *2011 Frontiers in education conference (FIE)*. IEEE, F1C–1.
- [38] Alex Lishinski, Aman Yadav, Jon Good, and Richard Enbody. 2016. Learning to Program: Gender Differences and Interactive Effects of Students’ Motivation, Goals, and Self-Efficacy on Performance. In *Proceedings of the 2016 ACM Conference on International Computing Education Research (ICER ’16)*. 211–220. <https://doi.org/10.1145/2960310.2960329>
- [39] Stephanie Lunn, Maira Marques Samary, Susanne Hambruch, and Aman Yadav. 2022. Forging a Path: Faculty Interviews on the Present and Future of Computer Science Education in the United States. *ACM Transactions on Computing Education (TOCE)* 22, 4 (2022), 1–24.
- [40] Jane Margolis and Allan Fisher. 2002. *Unlocking the clubhouse: Women in computing*. MIT press.
- [41] Paola Medel and Vahab Pournaghshband. 2017. Eliminating gender bias in computer science education materials. In *Proceedings of the 2017 ACM SIGCSE technical symposium on computer science education*. 411–416.
- [42] Kelly Miller, Gerhard Sonnert, and Philip Sadler. 2018. The influence of students’ participation in STEM competitions on their interest in STEM careers. *International Journal of Science Education, Part B* 8, 2 (2018), 95–114.
- [43] Kathryn C Oleson. 2023. *Promoting inclusive classroom dynamics in higher education: A research-based pedagogical guide for faculty*. Taylor & Francis.
- [44] Joan Peckham, Lisa L Harlow, David A Stuart, Barbara Silver, Helen Mederer, and Peter D Stephenson. 2007. Broadening participation in computing: issues and challenges. *ACM SIGCSE Bulletin* 39, 3 (2007), 9–13.
- [45] Yolanda A. Rankin and Jakita O. Thomas. 2019. Straighten Up and Fly Right: Rethinking Intersectionality in HCI Research. *Interactions* 26, 6 (2019), 64–68. <https://doi.org/10.1145/3363033>
- [46] Johnmarshall Reeve. 2009. Why teachers adopt a controlling motivating style toward students and how they can become more autonomy supportive. *Educational psychologist* 44, 3 (2009), 159–175.
- [47] Johnny Saldaña. 2021. *The Coding Manual for Qualitative Researchers* (4th ed.). SAGE Publications.
- [48] Linda J Sax, Kathleen J Lehman, Jerry A Jacobs, M Allison Kanny, Gloria Lim, Laura Monje-Paulson, and Hilary B Zimmerman. 2017. Anatomy of an enduring gender gap: The evolution of women’s participation in computer science. *The Journal of Higher Education* 88, 2 (2017), 258–293.
- [49] Kshitij Sharma, Juan C. Torrado, Javier Gómez, and Letizia Jaccheri. 2021. Improving girls’ perception of computer science as a viable career option through game playing and design: Lessons from a systematic literature review. *Entertainment Computing* 36 (2021), 100387. <https://doi.org/10.1016/j.entcom.2020.100387>
- [50] Alison Singer, Georgina Montgomery, and Shannon Schmoll. 2020. How to foster the formation of STEM identity: studying diversity in an authentic learning environment. *International Journal of STEM Education* 7 (2020), 1–12.
- [51] Jonathan Smith and Isabella Nizza. 2022. *Essentials of interpretative phenomenological analysis*. <https://doi.org/10.1037/0000259-000>
- [52] Jonathan Aa Smith. 2015. Interpretative phenomenological analysis as a useful methodology for research on the lived experience of pain. *British journal of pain* 9(1), 41–2 (2015), 2049–4637. <https://doi.org/10.1177/2049463714541642>
- [53] Bernadette Spieler, Libora Oates-Indruchová, and Wolfgang Slany. 2020. Female students in computer science education: Understanding stereotypes, negative impacts, and positive motivation. *Journal of Women and Minorities in Science and Engineering* 26, 5 (2020).
- [54] Claude M. Steele and Joshua Aronson. 1995. Stereotype Threat and the Intellectual Test Performance of African Americans. *Journal of Personality and Social Psychology* 69, 5 (1995), 797–811. <https://doi.org/10.1037/0022-3514.69.5.797>
- [55] Gijsbert Stoet and David C. Geary. 2018. The Gender-Equality Paradox in Science, Technology, Engineering, and Mathematics Education. *Psychological Science* 29, 4 (2018), 581–593. <https://doi.org/10.1177/0956797617741719>
- [56] Emily A Thrush and Linda Hooper. 2006. Industry and the academy: How team-teaching brings two worlds together. *Technical Communication* 53, 3 (2006), 308–316.

- [57] Christoph Treude and Hideaki Hata. 2023. She elicits requirements and he tests: Software engineering gender bias in large language models. In *2023 IEEE/ACM 20th International Conference on Mining Software Repositories (MSR)*. IEEE, 624–629.
- [58] Jamaica Vizcarra-Garcia et al. 2021. Teachers’ Perceptions of Gender Inclusive Language in the Classroom. *International Journal of Linguistics, Literature and Translation* 4, 3 (2021), 110–116.
- [59] Jennifer Wang, Hai Hong, Jason Ravitz, and Marielena Ivory. 2015. Gender differences in factors influencing pursuit of computer science and related fields. In *Proceedings of the 2015 ACM Conference on Innovation and Technology in Computer Science Education*. 117–122.
- [60] Susanne Wollin-Giering, Markus Hoffmann, Jonas Höfting, and Carla Ventzke. 2023. Automatic transcription of qualitative interviews. *Sociology of Science Discussion Papers* URL: https://www.static.tu-berlin/fileadmin/www/10005401/Publikationen_sos/Wollin-Giering_et_al_2023_Automatic_transcription.pdf (2023).
- [61] Ursula Wolz, Lina Battestilli, Bruce Maxwell, Susan Rodger, and Michelle Trim. 2018. Best practices in academia to remedy gender bias in tech. In *Proceedings of the 49th ACM Technical Symposium on Computer Science Education*. 672–673.
- [62] Megan Woods, Trena Paulus, David P. Atkins, and Rob Macklin. 2016. Advancing Qualitative Research Using Qualitative Data Analysis Software (QDAS)? Reviewing Potential Versus Practice in Published Studies using ATLAS.ti and NVivo, 1994–2013. *Social Science Computer Review* 34, 5 (2016), 597–617. <https://doi.org/10.1177/0894439315596311>

APPENDIX

A INTERVIEW GUIDE (SEMI-STRUCTURED)

Note on confidentiality and scope. To minimise re-identification risk, participants were asked to avoid naming individuals, specific unit codes, or incidents that could identify themselves, colleagues, students, or institutions. The interview guide focuses on educators' perspectives on *gender inclusivity*. While other identity dimensions may intersect with gender, this study does not conduct a layered intersectional analysis beyond gender due to limited and non-reportable data across additional dimensions.

The following questions were used as a semi-structured guide. Prompts were adapted as needed to follow participants' responses and to probe for clarification.

RQ1: Educators' perceptions of gender inclusivity in CSHE

- (1) In the context of your unit's teaching and learning, what does *gender inclusivity* mean to you in practice?
- (2) Could you share any teaching experiences related to gender inclusivity (or lack of gender inclusivity) in your unit(s)?
- (3) In your opinion, how does gender influence students' experiences within your classrooms or units, if at all?
- (4) Can you describe any challenges or barriers you have noticed for students of various genders in your classroom(s) or other teaching-related settings?
- (5) How do you think gender inclusivity might affect students' learning outcomes and overall learning experience?
- (6) What approaches can educators adopt to create a more inclusive and welcoming environment for students of all genders in computer science higher education?

RQ2: Factors influencing gender inclusivity in CSHE learning environments

- (1) How do you integrate gender inclusivity into your teaching practices?
- (2) In what ways do you identify needs that may differ across students of different genders in your classroom, and how (if at all) do you modify your teaching approach to accommodate those needs?
- (3) Can you elaborate on any teaching methods or strategies you have employed to foster gender inclusivity in your classroom(s)?
- (4) What challenges or barriers have you encountered while developing inclusive teaching, learning, and assessment materials or strategies?
- (5) How do you judge whether your inclusive teaching practices are working (e.g., based on student feedback, participation patterns, or other signals)?
- (6) Are there any resources or tools that you utilise or are aware of to support inclusive teaching and assessment?

RQ3: Educators' perceptions of student facets influencing participation and engagement

- (1) How would you define student participation in your classroom(s), and what are the primary indicators of engagement and involvement?
- (2) Can you identify any student characteristics, traits, or behaviours that may impact their involvement in learning activities?
- (3) What challenges or barriers might students encounter when participating in learning activities, and how do you address these issues (if at all)?